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Interaction design

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Undergraduate study in
Computing and related programmes

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This is one of a series of subject guides published by the University. We regret that due to pressure of work the author is unable to enter into any correspondence relating to, or arising from, the guide. If you have any comments on this subject guide, favourable or unfavourable, please use the form at the back of this guide.

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Preface

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About this course unit

CIS 348, Interaction design, is a half unit Level 3 option for the BSc/Diploma in Computing and Information Systems (CIS) and the BSc/Diploma in Creative Computing. The course has no specific or formal pre-requisite course unit or technical requirements. However, students should have a good general awareness and experience of a range of current computer-based systems, ranging from stand-alone and mobile applications to internet-based and commercial ones. To obtain the most benefit from this course unit, students should be interested in the human aspects of using new technologies and novel interaction media. The course unit does not cover applications programming, software development, technological solutions or analysis of such systems. An interest in and willingness to learn about aspects of user psychology, most especially some basic concepts in experimental and cognitive psychology, will be necessary. Although this may appear challenging for those with no prior experience in this area, the core Essential and Further reading should suffice.

CIS 348 is an updated replacement for CIS 315, Human Computer Interaction, and shares some basic material with it. Prior material on sample examination questions and past *Examiner's reports* will still be broadly applicable. The course unit has been brought up-to-date, its emphasis refocused more on the design of human-computer and human-technology interactions, based on the needs and requirements of the users of such technologies, and assessing how such interactive applications are used in real-life. This course unit introduces the study of Human Computer Interaction (HCI) together with the newer area of Interaction Design (ID).

Knowledge of interface design and user requirements is critical to many aspects of software development and applied computing. Increasingly, resources are devoted to the design and optimal functioning of interfaces to an increasing range of interactive technologies. Standards and accessibility legislation and guidance are now a crucial element in commercial interface design, which means that new applications are being created with an emphasis on good design and user participation. The world that HCI investigates and researches is developing into a mobile, ubiquitous and ambient one. The 'computer' has been extended and, in many cases, replaced by numerous mobile interactive devices. Users are no longer homogeneous but increasingly form new and distinct user groupings. The concept of a usable interface has been revolutionised by devices such as the iPod and iPhone, and people increasingly recognise and expect to be able to use electronic devices easily and without difficulty. The global expansion of technology means that localisation and internationalisation issues assume greater importance. The expansion of commercial web-based applications in e-commerce and e-government will undoubtedly continue apace. New types of interaction based on gesture, touch, motion and speech are increasingly common and can be seen in current commercial products. Social networking, community computing and immersive technologies have become a fundamental part of the lives of many, and home-based applications of new technologies abound. Games and the associated new markets in entertainment computing, learning technologies and virtual applications continue to push the boundaries of creative computing. An increasingly elderly population in many parts of the world has led to a new emphasis on assistive technologies and for health-related technology-based applications.

Given this snapshot of current impacts, students need to be able to engage with future technologies and to fully appreciate and understand the design principles which create effective interactive applications, and to be well-prepared to apply them in their future jobs and careers as computer professionals and practitioners. This introductory course unit is thus essential for anyone who will be involved in the analysis, design, development or evaluation of interactive applications of any kind. The aim is to go beyond mere 'guidelines for good design' and to enable students to develop an understanding of the principles of interaction, to identify problems with an interface, to understand how such interactions may be improved and to gain the knowledge necessary to be able to design better interfaces. This Study Guide will introduce and discuss the topics and technologies described in the previous paragraph and will enable students to gain an up-to-date understanding of what it means to be an Interaction Designer.

Assessment

Important: The information and advice given in the section below are based on the examination structure used at the time this guide was written. Please note that subject guides may be used for several years. Because of this we strongly advise you to always check the current *Regulations* for relevant information about the examination. You should also carefully check the rubric/instructions on the paper you actually sit and follow those instructions.

The course unit is assessed by an examination of three (out of a choice of five) questions. The examination questions are essay-based but do have a strong element of applied design and interpretation of the techniques studied during the course unit. Guidance is given in the next section. There are also coursework assignments. The essay- or report-type assignments are similar to those that have recently been set for the CIS 315 course unit and will change in topic every year. They should allow students to become better informed about selected specific aspects of Human Computer Interaction and Interaction Design and to enable exploration of the extensive online information available for this subject. Again, specific details and guidance can be found later in the Study Guide.

Examination guidance

For appropriate examination preparation, please take note of the following points:

- Take a few minutes to make a plan of each question and to gather your thoughts instead of immediately starting to write.
- Ensure that you fully understand the topic area of the question.
- Ensure that you can answer every part and section of the question. Being able to answer only part of the question will not help you achieve a good overall mark.
- Do not spend unnecessary time restating the question, either in your own words or in repeating the question text. This is not required and will use up valuable time.
- Do not spend examination time answering one question at the expense of others: it is generally better to answer three questions in full than one in great detail and two very briefly.
- Do not repeat details from one section in another section: it is unlikely that this is what the Examiner intended and the focus of your answer in each section should be quite distinct.
- Ensure that the level and detail of the answer you give corresponds to the marks allocated. Do not spend too much time and effort on a part of the question that is worth only a few marks. Similarly, do not merely write cryptic notes or single points for a part of the question that is worth a large number of marks. Try to achieve the balance reflected in the marks indicated.
- Read the question carefully and answer in the way that is requested: wording such as ‘describe’, ‘compare and contrast’, ‘itemise’, ‘illustrate’, ‘explain with diagrams’ tells you what sort of answer is expected and what sort of detail you should go into. Make sure you understand what type of answer is required and do provide diagrams or examples where requested as this is part of the marking scheme for the question.
- Do not spend time providing unnecessary diagrams. If diagrams are requested, they should be clearly labelled and described. Providing as an answer an unlabelled diagram from memory without a description will not attract good marks.
- Do try to use tables and lists where appropriate – for example, in a question which asks you to contrast two approaches or itemise the differences between two aspects of a topic.
- Organise your time to allow sufficient time to read over what you have written and to make any necessary corrections.

Coursework guidance

Coursework will generally be in an essay format. A detailed statement of what is required will be produced, with the marking scheme clearly indicated and suitable references given. The following guidance will help you to produce high quality coursework:

- You will be required to provide an electronic copy for plagiarism checking purposes. Any submission without any electronic copy will not be marked. Very short submissions are also unacceptable.
- You should write in a report or essay format – not in note or bullet-point form – with a defined structure as detailed in the coursework instructions. You do not need to restate the question asked or provide a

table of contents, an index or a cover sheet, or any extra information or appendices, and you should attempt to minimise the use of paper.

- The structure, clarity and organisation of your work will be assessed and some marks awarded for it. Your submission must be well-presented in a coherent and logical fashion. It should be spell- and grammar-checked and you should structure it so that you have both a clear introduction and a conclusion. A concluding section is a required part of your coursework submission.
- You should include relevant diagrams, drawings, illustrations or images such as graphics and screenshots as coursework is likely to involve a design scenario and production of paper-based prototypes. These will have an impact on the readability and presentation value of your work.
- Remember that the focus of coursework will normally be on HCI aspects and on design principles for user interactions, not about applications as such, the hardware and software involved, or business elements. You should be assessing how humans use interactive applications and what the elements of a good user-computer interface are.
- You must provide a Bibliography and References section at the end of your work, showing the books, articles and websites you have referenced and consulted. All books cited, reports referred to and any material used (including all online resources) must be referenced. Students who do not provide references, or cite only the course textbook and Study Guide, will be marked down.
- Websites should be referenced by the date of access and a complete and correct URL (generic site names are not acceptable). Other references should be in a standard format (i.e. author names (correctly spelled), year of publication, title, publisher, actual page numbers referenced). For a useful guide to referencing procedure, look at: http://education.exeter.ac.uk/dll/studyskills/harvard_referencing.htm
- The submitted assignment must be your own individual work and must not duplicate another person's or an author's work. Copying, plagiarism and unaccredited and wholesale reproduction of material from textbooks or from any online source are unacceptable. Any such work will be severely penalised or rejected. Any text that is not your own words and which is taken from any source must be placed in quotation marks and the source identified correctly in the References section. Failure to do so will incur very serious penalties.
- Do be extremely careful about the validity of information on internet sites and world wide web sources. Citing and copying from Wikipedia and other encyclopaedic sources, for example, is not appropriate for a coursework submission. Be aware that many information sites are commercial advertising, or simply repeat material from elsewhere. Do check the date of all material and do not use out-of-date sites, sites which list student work or projects, references from commercial publishers to journal paper abstracts only or those which are simply personal opinions, blogs or comments. Please be selective and critical in your choice of material.

Course aims

This course unit provides an understanding of the theoretical and methodological issues that can be applied to the design and evaluation of interactive computer-based systems.

Learning objectives

Students should, at the end of the course unit, understand a broad range of basic concepts in HCI and ID, and be able to apply HCI principles, guidelines and techniques to the analysis, design and evaluation of interactive systems, no matter what the underlying technology.

Learning outcomes

By the end of this half unit, and having completed the relevant readings and activities, you should be able to:

- gain a historical perspective of the field of HCI and its relationship with software engineering, psychology and ergonomics
- understand what Interaction Design is and be able to think critically about design
- appreciate HCI theory, practice and a set of design approaches
- understand the concept of usability and techniques of usability evaluation
- make use of usability concepts in appreciating, using and building interactive solutions for a range of applications
- criticise and improve poor interface designs, and so develop the skills to design and evaluate interfaces
- demonstrate awareness of new application areas and up-and-coming advanced technologies in order to better understand the potential of new technologies and techniques in the design of future interactive systems and applications.

The Study Guide

This course unit offers a multidisciplinary introduction to interaction with technology (and not just human computer interaction), concentrating on the design concepts which underlie current Interaction Design. This course will provide a basic-level grounding in the knowledge and skills applicable to the specification, design, prototyping and evaluation of advanced interactive systems. These tools and methods are used by software developers, user experience and interaction designers, designers of applications for mobile and hand-held devices, games developers and usability and Human Factors specialists.

How to use the Study Guide

This Study Guide is not a course text but sets out topics for study in the CIS 348 course unit. Additional background material is provided where applicable and links to useful information sources and examples (such as case studies, online examples, video material, and downloadable e-books and PDFs) are given. Essential reading is listed at the start of each chapter; Further reading and websites at the end. There is a recommended course textbook, with supporting online material, linked to the topics in this course unit. Required reading is identified and linked to specific chapters of this textbook; it should be undertaken along with the study of each topic and chapter in this Study Guide. Further personal reading and study to enhance overall understanding of the subject is an essential part of this course unit. Additional recommended readings are provided, full details of which are given at the end of each chapter and in Appendix 2: Bibliography. The Study Guide is intended to support and reinforce this personal study, not to supplant it. It will not be possible to pass this course unit by reading only the Study Guide.

Structure of the guide

a. Chapters

- Chapter 1, **Introduction to HCI and Interaction Design**, introduces HCI and ID and positions them both within surrounding disciplines. A brief history of the field, showing its genesis and development towards a design discipline, is given.
- Chapter 2, **Interaction and design approaches**, provides an overview of design approaches, with examples and descriptions.
- Chapter 3, **Techniques for Interaction Design requirements**, details selected specific approaches and techniques for studying users and carrying out user-centred task design and user analysis.
- Chapter 4, **Usability**, gives an explanation of usability requirements and **usability engineering**.
- Chapter 5, **Evaluation and usability assessment techniques**, details some usability evaluation and assessment strategies and techniques.
- Chapter 6, **Designing for different users**, considers design guidelines and standards, accessibility requirements, and issues involved in designing for specific populations. This encompasses globalisation and internationalisation aspects.
- Chapter 7, **Design case studies**, is devoted to case study material.
- Chapter 8, **Interaction Design and new technologies**, considers current Interaction Design questions and approaches for new and emerging technologies and paradigms.
- Chapter 9, **Real world interactions**, follows on by considering real-world applications and sectors where HCI and ID are having a wider societal impact, such as healthcare, collaborative and communicative technologies, and the developing world.

[1] These quotations are personal communications to the Study Guide author and cannot be cited.

Each chapter is introduced by a pertinent quotation from an HCI 'guru'[1] and is preceded by a tag cloud and a graphical roadmap outline of that chapter. The main text is organised in numbered sections. Each chapter concludes with a summary of the main points and a statement detailing the learning objectives for that topic. Each chapter indicates the essential reading required for full comprehension of the topic.

To assist with revision, Appendix 3 at the end of the Study Guide contains the roadmap, the summary and the learning objectives for each chapter. There is no Summary in the Preface.

b. Essential reading, references and footnotes

Essential reading will normally be a chapter in the designated course unit textbook, or one of the alternative, older textbooks. Additional references to other material and recommended Further reading can be found at the end of a chapter. These have been chosen to enhance knowledge and expand coverage. They are usually books or free-access websites. The Bibliography appendix has the full citable details (and, where available, the online link to the publisher's website page for that book) of all the textbooks and print sources referred to in the Study Guide. At the end of a chapter, the recommended reading is referenced in a short format only, for example:

Surname, Initial (Date of publication). Book title. Chapter or page numbers to look at. URL.

Footnotes are usually citation details for quotations or for particular articles or websites referred to. These additional links and citations can be followed up, if wished, but it is not mandatory: they do not necessarily appear in the bibliography appendix.

c. Exercises and Sample examination questions

Themed short questions, suggested exercises and discussion points, together with sample examination-type questions, are provided to facilitate revision and to test understanding of the concepts covered in that chapter. Local tutors may also wish to use these to lead discussion groups, so model answers are not provided. A Sample examination paper, together with possible answers in note form, can be found at the end of the Study Guide.

Suggested study time

The *Student Handbook* states the following: 'To be able to gain the most benefit from the programme, it is likely that you will have to spend at least 250 hours studying for each full unit, though you are likely to benefit from spending up to twice this time'. Note that this subject is a half unit.

It is suggested that a chapter be read in detail, and immediate notes taken, each week. At the same time, the associated readings and web-based material should be accessed and noted. It is advisable to attempt the practice questions and to access relevant case study or video material when studying a particular topic. Some of these may be design exercises involving paper and pen prototyping, so sufficient time should be set aside, depending on how many are chosen. Revision should take place over a number of weeks before the examination, and practice examination questions undertaken on a timed basis.

Recommended texts

The recommended textbook for this course unit is an updated and revised edition of a long-standing text.

> Shneiderman, B., C. Plaisant, M. Cohen and S. Jacobs *Designing the user interface: strategies for effective human-computer interaction*. (Boston: Addison-Wesley, available from Pearson Education, 2009) [ISBN 0321537351; 9780321537355] fifth edition. Price in 2010: £40–80. www.pearsonhighered.com/dtui5einfo/

The book includes much practical advice, research-supported design guidance and instructions for starting a first HCI design project. Each chapter begins with coverage detail and a chapter outline and concludes with a 'Practitioner's Summary' and a 'Researcher's Agenda' giving contrasting or complementary views of the topic covered. A useful companion website (http://wps.aw.com/aw_shneiderman_dtui_5/) comprises a blog written by one of the co-authors, self-assessment quizzes, discussion questions, projects, useful general web links, links to available online case studies and video material, and copies of PowerPoint presentations allied to the text of the book. Some of this text is free to view, other parts of the site are restricted access and available only to those who purchase the textbook and associated access codes. This edition of the textbook is the most up-to-date one; previous editions are not suitable since there is much new material. If this book is not available then the two alternative textbooks indicated below (those used for CIS 315) may be used – but do make use of the online resources for this textbook.

Three additional textbooks are recommended, but are not mandatory. It should be possible to access them from a library, or to share between a number of students. Again, these are recently published texts and it is unlikely that many used copies will be available until after 2011.

1. Lazar, J., H.F. Jinjuan and H. Hochheiser *Research methods in human computer interaction*. (John Wiley and Sons, 2009) [ISBN 9780470723371; 0470723378]. Price in 2010: £34.99. http://eu.wiley.com/WileyCDA/WileyTitle/productCd-0470723378.html

This book concentrates on the practicalities of carrying out user research and testing and gives very specific detailed advice and information on the whole area of research methodology. References will be made to this textbook at appropriate points in the Study Guide. It might also be a useful purchase for those considering a design-based, HCI or ID flavoured third year project. Note that another textbook with the title 'Research Methods for Human Computer Interaction', edited by P. Cairns and A. Cox, exists; this is not the textbook which is recommended but is an edited collection of essays on research methods more suitable for students with a background in social sciences. It is suggested as an alternative, if necessary.

2. Norman, K.L. *Cyberpsychology: an introduction to human-computer interaction*. (Cambridge University Press, 2008) [ISBN 0521687020; 9780521687027]. Price in 2010: £24.99. www.cup.cam.ac.uk/us/catalogue/catalogue.asp?isbn=9780521867382

Kent Norman's book concerns the psychology of how humans and computers interact. It contains a brief history of psychology and computers and an introduction to theories and models of human–computer interaction, covering relevant topics in sensation and perception, learning and memory, individual differences, motivation and emotion and social relations. Some specific issues in assistive technologies, video games, and electronic education are also discussed. Reference will be made to this textbook at appropriate points in the Study Guide.

3a. Buxton, B. *Sketching user experience: getting the design right and the right design*. (Morgan Kaufman, 2007) [ISBN 0123740371; 9780123740373]. Price in 2010: £29.99. www.elsevier.com/wps/find/bookdescription.cws_home/711463/description#description

3b. Buxton, B., S. Greenberg and S. Carpendale *Sketching user experience: the workbook*. (Morgan Kaufman, 2010) [ISBN 9780123819598]. Price in 2010: £12.99. www.elsevier.com/wps/find/bookdescription.cws_home/723098/description#description

Bill Buxton's book (and the associated workbook) is a masterclass in design, which helps non-designers understand and become practised in the skills of sketching, prototyping, creating mock-ups and interactive modelling which underlie much of Interaction Design. Various techniques and strategies are explained and opportunities given for personal practice which should lead to a better understanding of storytelling, critiquing and evaluating designs. This book is strongly recommended for Chapter 2 and for many of the exercises, especially those concerned with design and prototyping. It would be invaluable for those students committed to carrying out a design-based project. Other similar books do exist and some are referenced in Chapter 2 but Buxton's book is the only one with a workbook specifically linked to it. Note that there are, at the time of writing, no online support materials available.

Links to alternative texts (associated with course unit CIS 315)

Dix, A., J. Finlay, G. Abowd and R. Beale *Human-computer interaction*. (Prentice Hall, 2006) [ISBN 0130461091; 9780130461094] third edition. Price in 2010: £30–65. www.hcibook.com/e3/

Sharp, H., Y. Rogers and J. Preece *Interaction design: beyond human computer interaction*. (John Wiley and Sons, 2007) [ISBN 0470018666; 9780470018668] second edition. Price in 2010: £20–40. www.id-book.com/

The two textbooks which were recommended for the previous course unit CIS 315 are now increasingly outdated, even though both have been published in new editions and future editions are currently being prepared. The original and first issues of both textbooks are not at all suitable as both are very much out of date. Both textbooks now have associated websites and teaching resources. The former has an associated set of PowerPoint presentations and online examples and questions (www.hcibook.com/e3/) whilst the latter is especially valuable for its case studies (www.id-book.com/casestudy_index.htm) and for the links associated with chapters and interactive educational tools (www.id-book.com/interactivities_index.htm). References in both of these texts will be listed under essential reading material at the end of relevant chapters in this Study Guide.

Supporting resources

Each of the three previously listed HCI textbooks has a set of supporting extra resources, accessed from dedicated websites. A brief listing is given below of the chapter headings and support material in Shneiderman et al. (2009).

Recommended textbook

Shneiderman et al. (2009).

Each chapter ends with a 'World Wide Web Resources' section at the companion website. http://wps.aw.com/aw_shneiderman_dtui_5/

Part I: Introduction

1. Usability of Interactive Systems
2. Guidelines, Principles, and Theories

Part II: Development Processes

3. Managing Design Processes
4. Evaluating Interface Designs

Part III: Interaction Styles

5. Direct Manipulation and Virtual Environments
6. Menu Selection, Form Fill-in, and Dialog Boxes
7. Command and Natural Languages
8. Interaction Devices
9. Collaboration and Social Media Participation

Part IV: Design Issues

10. Quality of Service
11. Balancing Function and Fashion
12. User Documentation and Online Help
13. Information Search
14. Information Visualization

Afterword: Societal and Individual Impact of User Interfaces

Each chapter ends with a 'World Wide Web Resources' section at the Companion website: http://wps.aw.com/aw_shneiderman_dtui_5/

The Open Reader Resources http://wps.aw.com/aw_shneiderman_dtui_5/110/28381/7265752.cw/index.html include:

- A blog composed by co-author Steve Jacobs
- Additional discussion questions linked to each chapter
- Links to topics mentioned in each chapter
- User Interface Video Links which point to 40+ videos from past to present, and which link to video libraries.

The Protected Reader Resources are available to those who purchase the textbook and contain:

- Self-assessment quizzes
- Powerpoint slides for all chapters
- Project topics in 'Rapid Prototyping', 'Serving the Users' Needs', 'Information Visualization' and 'HCI Theories'.

Alternative textbook 1

Sharp et al. (2007)

Selected Interaction Design resources under the heading of 'Starter' ranging from blogs to software recommendations. www.id-book.com/starters.htm

Chapter headings, with links to each chapter's introduction and aims. www.id-book.com/chapter_index.htm

1. What is Interaction Design?
2. Understanding and Conceptualizing Interaction
3. Understanding Users
4. Designing for Collaboration and Communication
5. Affective Aspects
6. Interfaces and Interactions
7. Data Gathering
8. Data Analysis, Interpretation and Presentation
9. The Process of Interaction Design
10. Identifying Needs and Establishing Requirements
11. Design, Prototyping and Construction
12. Introducing Evaluation
13. An Evaluation Framework
14. Usability Testing and Field Studies
15. Analytical Evaluation

- Web resources which extend the subject treatment in each chapter consist mainly of further readings, links to blogs, presentations and slideshows (all were current in 2006/7 but have not been updated).
- Comments on the relevant assignments set at the end of each chapter.
- A set of teaching materials as PowerPoint slides based on the content of each chapter content. www.id-book.com/slides_index.htm
- Supporting case studies which illustrate the practice of Interaction Design. They are summarised in the book, but the online files contain more detailed descriptions and additional visual material. www.id-book.com/casestudy_index.htm
- A section entitled **Interactivities** which contains downloadable interactive educational tools. www.id-book.com/interactivities_index.htm

Alternative textbook 2

Dix et al. (2006) www.hcibook.com/e3/

Chapter headings, with a chapter outline, exercises and links to more specific topics (again, current in 2006 and not all available). www.hcibook.com/e3/chapters/intro/

Part One: Foundations

1. The human
2. The computer
3. The interaction
4. Paradigms

Part Two: Design Process

5. Interaction design basics
6. HCI in the software process
7. Design rules

8. Implementation support
9. Evaluation techniques
10. Universal design
11. User support

Part Three: Models and Theories

12. Cognitive models
13. Socio-organizational issues and stakeholder requirements
14. Communication and collaboration models
15. Task Analysis
16. Dialogue notations and design
17. Models of the system
18. Modelling rich interaction

Part Four: Outside the Box

19. Groupware
20. Ubiquitous Computing and augmented realities
21. Hypertext, multimedia and the World Wide Web

- Exercises and solutions relevant to each chapter. www.hcibook.com/e3/exercises/
- Slides available in PowerPoint format or as student handouts. www.hcibook.com/e3/resources/
- Additional online only extra material which is linked to specific pages and sections in the book, or to certain exercises. Further links lead to relevant other material and sites. www.hcibook.com/e3/online/
- Case studies. www.hcibook.com/e3/casestudy

Sources of further information

A very wide range of HCI and ID material is available online. HCI-related search terms will provide a plethora of potential sources, some more reliable and trustworthy than others. The lists below are selected sources that are deemed pertinent and useful but such a list is not exhaustive and is very time-sensitive. There is also an extensive annotated list of resources in Shneiderman et al. (2009) Chapter 1, pp.42–53.

Glossaries of terms

Glossaries are available online and elsewhere:

- Glossary of terms in: Lindgaard, G. *Usability testing and system evaluation: a guide for designing useful computer systems.* (London: Chapman and Hall, 1994) [ISBN 0412461005].
- SAP usability glossary. www.sapdesignguild.org/resources/glossary_usab/
- Usability first glossary. www.usabilityfirst.com/glossary/main.cgi

HCI, ID and usability interest groups

- ACM Special Interest Group on Computer Human Interaction (SIGCHI). www.sigchi.org/
- interaction (formerly known as The British HCI Group). www.bcs.org/

server.php?show=nav.14296

- Interaction Design Association (IxDA). A global network dedicated to the professional practice of Interaction Design. www.ixda.org/
- The Information Architecture Institute (IAI). Supports individuals and organisations specialising in the design and construction of shared information environments. www.iainstitute.org/
- The Usability Professionals Association (UPA). www.upassoc.org/

Magazines

- Interfaces (BCS interaction Specialist Group). www.bcs-hci.org.uk/about/interfaces/archive
- Interactions (ACM SIGCHI). http://interactions.acm.org/
- User Experience: The Magazine of the Usability Professionals' Association. www.upassoc.org/upa_publications/user_experience/ (subscription)
- Communications of the ACM (Association for Computing Machinery). http://cacm.acm.org/magazines/
- Computer Journal (British Computer Society). http://comjnl.oxfordjournals.org/; http://comjnl.oxfordjournals.org/archive/ (subscription)

Journals

- Behaviour & Information Technology (Taylor and Francis). www.tandf.co.uk/journals/tf/0144929X.html
- Cognition, Technology and Work (Springer). www.springer.com/computer/journal/10111
- Computer Supported Cooperative Work (Springer). www.springer.com/new+%26+forthcoming+titles+(default)/journal/10606
- Computers in Human Behavior (Elsevier). www.elsevier.com/wps/find/journaldescription.cws_home/759/description
- CyberPsychology and Behavior (Liebert). www.liebertpub.com/products/product.aspx?pid=10
- Entertainment Computing (Elsevier). www.elsevier.com/wps/find/journaldescription.cws_home/717010/description#description
- Foundations and Trends in Human-Computer Interaction (NOW). www.nowpublishers.com
- Human-Computer Interaction (LEA). http://hci-journal.com
- Interacting with Computers (Elsevier). www.elsevier.com/wps/find/journaldescription.cws_home/525445/description#description
- International Journal of Human-Computer Studies (Elsevier). www.elsevier.com/wps/find/journaldescription.cws_home/622846/description#description
- International Journal of Mobile Human Computer Interaction (IGI). www.igi-global.com/journals/details.asp?id=8050
- Online International Journal of Usability Studies (UPA). www.upassoc.org/upa_publications/jus/jus_home.html
- Personal and Ubiquitous Computing (Springer). www.springer.com/computer/hci/journal/779
- Pervasive and Mobile Computing (Elsevier). www.elsevier.com/wps/find/

journaldescription.cws_home/704220/description#description

- Transactions on Accessible Computing (ACM Press). www.is.umbc.edu/taccess/index.html
- Transactions on Affective Computing (IEEE). www.computer.org/portal/web/tac
- Transactions on Computer Human Interaction/TOCHI (ACM Press). http://tochi.acm.org/charter.shtml
- Transactions on Human-Computer Interaction (AIS). http://aisel.aisnet.org/thci/
- Universal Access in the Information Society (Springer). www.springerlink.com/content/107725/
- Virtual Reality (Springer). www.springer.com/computer/computer+processing/journal/10055

Conference Proceedings

- Human Factors in Computer Systems, Proceedings of the ACM SIGCHI Conferences, Special Issue of the SIGCHI Bulletin. (ACM Press, 1983 to present). www.sigchi.org/conferences, www.interaction-design.org/references/conferences/series/chi.html
- People and Computers, Proceedings of the HCI conferences of the BCS HCI Specialist Group. (Cambridge University Press; Springer; other publishers, 1984 to present). www.interaction-design.org/references/conferences/series/bcshci_people_and_computers.html
- Proceedings of Interact Human-Computer Interaction Conference. (North-Holland; Springer; other publishers, 1984 to present). www.interaction-design.org/references/conferences/series/interact.html
- Proceedings of some ACM SIGCHI sponsored specialised conferences. (ACM Press, various dates).
- ASSETS (conference on Assistive Technologies)
- CSCW (Computer Supported Cooperative Work)
- DIS (Designing Interactive Systems)
- Hypertext (Hypertext and Hypermedia)
- ITS (Interactive Tabletops and Surfaces)
- IUI (Intelligent User Interfaces)
- NordiCHI (Nordic Conference on Human-Computer Interaction)
- UbiComp (Ubiquitous Computing)
- UCMEDIA (User Centric Media)
- UIST (User Interface Software and Technology)
- Proceedings of conferences in associated areas (various publishers)
- ACE (Advances in Computer Entertainment Technology)
- ECSCW (European Conference On Computer Supported Cooperative Work) www.ecscw.org
- ETRA (Eye Tracking Research and Application)
- HAPTICS (The Haptics Symposium)

- IDC (Interaction Design and Children)
- INTETAIN (Intelligent Technologies for Interactive Entertainment)
- ISWC (IEEE International Symposium on Wearable Computers)
- IWEC (International Workshop on Entertainment Computing)
- MobileHCI (Human-Computer Interaction with Mobile Devices and Services)
- PDC (Participatory Design Conference)
- Pervasive (Pervasive Computing)
- TEI (Tangible, Embedded, and Embodied Interaction)
- UMAP (User Modelling, Adaptation and Personalization)
- UX (User Experience)

Web-based and online resources

It is notoriously difficult to maintain correct and up-to-date address details for web-based material, so the list below is presented with the caveat that it was checked and available at the time of writing, but continued correctness cannot be guaranteed. The list below contains a number of useful websites and resources for information in HCI, ID and Usability. Some are personal blogs, some commercial organisational sites, some are research community resources, some are quasi-official. View all with a critical eye.

Web resources

Blogs/Personal commentaries

- Alertbox www.useit.com/alertbox/
- Don Norman www.jnd.org/
- Johnny Holland http://johnnyholland.org/
- The Cooper Journal www.cooper.com/journal/

Commercial organisations/Company sites

- AM+A http://amanda.com/
- Fidelity Investments http://fcat.fidelity.com
- Putting people first www.experientia.com/blog
- SAP www.sapdesignguild.org/resources/resources.asp
- STC www.stcsig.org/usability/topics/index-alpha.html
- Foviance www.theusabilitycompany.com
- Usability First www.usabilityfirst.com/about-usability
- Usabilitynet www.usabilitynet.org/home.htm
- Usability Partners www.usabilitypartners.se/usability/links.shtml
- Usernomics www.usernomics.com/index.html
- Webcredible www.webcredible.co.uk/user-friendly-resources/
- WQUsability www.wqusability.com/articles/more-than-ease-of-use.html

Government/official sites

- Usability.gov www.usability.gov/index.html
- Web Access Initiative www.w3.org/WAI/

Newsfeeds and information dissemination

- Interaction Design www.interaction-design.org/
- Usability News www.usabilitynews.com/
- SURL Usability News http://psychology.wichita.edu/surl/usabilitynews

Research Community Resources

- HCC Education DL http://hcc.cc.gatech.edu/
- HCI Bibliography www.hcibib.org/
- HCI Index http://degraaff.org/hci/
- HCI Resource Net www.hcirn.com/
- The UX Bookmark www.theuxbookmark.com/
- UPA www.upassoc.org/usability_resources/
- Usable Web http://usableweb.com/
- UX Matters www.uxmatters.com/

Design related sites

- A List Apart www.alistapart.com/
- boxes and arrows http://boxesandarrows.com/
- Design Academy www.coolhomepages.com/cda/usability/
- Digital Web www.digital-web.com/
- Interaction Design Association www.ixda.org/resources
- Infodesign www.infodesign.com.au/usabilityresources
- Just Ask www.uiaccess.com/accessucd/index.html

Video material (libraries)

- Bill Buxton's Research Videos
  www.billbuxton.com/buxtonVideos.html
- HCC Education Digital Library
  http://hcc.cc.gatech.edu/taxonomy/videos.php
- Matthias Rauterberg's Videos in User-System Interaction
  www.idemployee.id.tue.nl/g.w.m.rauterberg/videos.html
- The Open Video Project
  www.open-video.org/collection_detail.php?cid=18

Special collections

- CSCW Video Special Collection
  www.open-video.org/featured_video.php?type=Special&cid=18
- SIGGRAPH Video Special Collection
  www.open-video.org/featured_video.php?type=Special&cid=19
- UIST Video Special Collection
  www.open-video.org/featured_video.php?type=Special&cid=20
- UBICOMP Video Special Collection
  www.open-video.org/featured_video.php?type=Special&cid=21
- CHI Video Retrospective Special Collection
  www.open-video.org/featured_video.php?type=Special&cid=8
- University of Calgary, Grouplab Videos
  http://grouplab.cpsc.ucalgary.ca/Videos

- University of Dundee, Computing Department Inclusive Digital Economy Network
  www.iden.org.uk/videoplayer.asp
- UTOPIA Project (Usable Technology for Older People: Inclusive and Appropriate)
  www.computing.dundee.ac.uk/projects/utopia/utopiavideo.asp
- University of Maryland Human Computer Interaction Laboratory (HCIL) library of videos
  www.cs.umd.edu/hcil/pubs/video-reports.shtml
- YouTube (usability)
  www.youtube.com/results?search_query=usability
- YouTube (Interaction Design)
  www.youtube.com/results?search_query=interaction+designandsearch=Search

Case studies

See Chapter 7 for the main list and other specific chapters for selected topics.

Acronyms

AH	Augmented Human
AI	Artificial Intelligence
AmbI/AmI	Ambient Intelligence
ANS	Autonomic Nervous System
ANSI	American National Standards Institute
BSI	British Standards Institute
CAD/CAM	Computer Aided Design/Computer Assisted Manufacturing
CBT/CAI	Computer Based Training/Computer Aided Instruction
CHI	Computer–Human Interaction
CSCW	Computer Supported Cooperative Work
DIN	*Deutsches Institut für Normung eV*
EIP	Exchanging Information with the Public
GOMS	Goals, Operators, Methods and Selection rules
HCI	Human Computer Interaction
IA	Information Architecture
ID	Interaction Design
IEC	International Electrotechnical Commission
ISO	International Standards Organisation
IT	Information Technology
KLM	Keystroke Level Model
MHP	Model Human Processor
MWBP	Mobile Web Best Practices
PD	Participatory Design
RFID	Radio Frequency Identification
RNIB	Royal National Institute for the Blind
SE	Software Engineering
TUI	Tangible User Interaction/Interfaces
Ubicomp	Ubiquitous Computing
UCD	User-Centred Design

UCSD	User-Centred System Design
UPA	Usability Professionals' Association
UX	User Experience
VDT	Visual Display Terminal
VE	Virtual Environment
VR	Virtual Reality
VRML	Virtual Reality Modelling Language
W3C	World Wide Web Consortium
WAI	Web Accessibility Initiative
WCAG	Web Content Accessibility Guidelines
WOZ	Wizard of Oz
WWW	World Wide Web
WYSIWYG	'What You See Is What You Get'

A picture is worth a thousand words. An interface is worth a thousand pictures.

Ben Shneiderman

applications cognitive computer design development engineering hci human information input interaction methods people research social systems techniques technology usability user

Chapter 1

Introduction to HCI and Interaction Design

Introduction » Definitions » Why study user interaction? » Early HCI » Why it changed » HCI in the 1970s and 1980s » The situation today » Summary

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- understand the definitions of HCI and ID and associated terminology
- demonstrate knowledge of the historical development of the discipline
- define and describe common HCI terms
- describe how future developments in interaction, design and technology link to the history of the study of ID and HCI.

Essential reading

Note: Further reading and websites are listed at the end of the chapter.

HCI textbooks

Shneiderman, B., et al. *Designing the user interface*. (2009) Part 1.

Sharp, H. et al. *Interaction design*. (2007) Chapter 3.

Dix, A., et al. *Human-computer interaction*. (2006) Chapter 1.

Introduction

A recent Microsoft research report entitled 'Being Human', takes as its starting point what life might be like in the near future:

> *What will our world be like in 2020? Digital technologies will continue to proliferate, enabling ever more ways of changing how we live. But will such developments improve the quality of life, empower us and make us feel safer, happier and more connected? Or will living with technology make it more tiresome, frustrating, angst-ridden, and security-driven? What will it mean to be human when everything we do is supported or augmented by technology?*[1]

[1] *Harper, R., T. Rodden, Y. Rogers and A. Sellen* Being Human: HCI in the Year 2020. *(Microsoft Research, 2008) [ISBN 978095547612].*

The aim of this report was to assess the changing interactive world and attempt to come to a better understanding of human relationships with advanced technologies. Some predictions are positive, some negative. On the one hand lifestyle changes assisted by technology can be extremely beneficial to health and well-being; digital media and tools can stimulate creativity; the ever-expanding world wide web enables immediate access to ever-growing information and data sources. On the other hand:

> *…governments become more reliant on computers to control society, criminals become more cunning via digital means and people worry more about what information is stored about them.*[2]

[2] *Harper, R. et al. (2008) op. cit.*

Why should this be important? The ways in which humans interact with computers, technology and devices continues to evolve rapidly and no end can be realistically envisaged. Ubiquitous Computing based on miniaturisation, portability and new display technologies is a reality; innovations in input modalities have brought about new forms of education and entertainment, and communication between people has been revolutionised because of unprecedented advances in network and communications technologies. The widespread use of computers and a vast range of interactive devices in commerce and industry means that we do now really live digital lives in a digital age. Wider access by those previously not part of the computer revolution has led to global social concerns – the impacts are all around us. To be able to engage in and change our future, it is critical to understand what those impacts are and how they can be mediated.

1.1 Definitions

Human Computer Interaction investigates the interplay between a human user and a computer system or interactive device through the medium of a user interface. The joint performance of tasks by humans and computers depends on human capabilities to use machines on the one hand, and the design (and design trade-offs) and implementation of interfaces on the other hand. HCI research concentrates on user needs and assesses interface designs and implementations according to usability criteria – it has science, engineering, and design aspects.

Interaction Design makes use of novel techniques and technologies to design, build, test and produce interactive applications. Interface, Interaction and User Experience Designers apply usability design principles to create productive, usable and enjoyable systems – and more satisfied users.

1.2 Why study user interaction?

Knowledge of Interaction Design, HCI and usability is increasingly recognised as a business requirement and design is now an integral part of the computer business, not least because of international standardisation and accessibility regulations. As the potential user population grows in size and diversity and computers become ambient, pervasive, ubiquitous and increasingly 'invisible', there is both an expanding awareness amongst users of what can be achieved and a blindness to the idea of the computer as a distinct separate device. Interaction and HCI design and research try to accommodate human diversity in cultural and international locations. As a basic tenet of usability, people deserve systems that are easy to learn and easy to use even though they have different physical and cognitive abilities.

Poor design choices are all too common. We need to understand why disasters, accidents and frustrations happen. Many safety and life-critical systems depend on computer control, including air traffic control; nuclear power plants; manned spacecraft; emergency services and medical instrument monitoring. There have been many documented critical failures due to identified design problems. Knowledge of how to design for human needs can, however, help minimise such errors and enable computer systems to adapt to user requirements, rather than the human having to fit to the demands of the technology.

1.3 Early HCI

HCI is about understanding and designing the relationships between people and computers. As a field of study, it is an amalgam of several scientific disciplines since the early 'man-computer symbiosis' suggested by Licklider in 1960.[3] The human side of HCI derived from physiology and applied psychology, and in particular, from ergonomics (an applied science with close ties to engineering and industrial applications). Ergonomics is essentially the design of equipment so that its operation is within the capacities of the majority of people. Human factors is similar but stems from the problems of designing equipment operable by humans within the limitation of sensory-motor features (e.g. the design of flight displays and command-and-control applications). In the 1950s ideas from communications engineering, linguistics, and computing led to a more experimentally-oriented discipline concerned with human information processing and performance. Human operation of computers was a natural extension – there were new problems with cognitive, communication, and interaction aspects not previously considered. The first applications of this research were to large-scale process control systems with mainframe computers and in military systems, with restricted input and output modalities. At the same time, cognitive psychologists concentrated on the learning of these and early Computer Based Training/Computer Aided Instruction systems, the transfer of that learning, mental representations, and human performance.

[3] *Licklider, J.C.R.* 'Man-computer symbiosis. IRE Transactions on Human Factors in Electronics', *HFE 1, 1960, pp.4–11.*

The advent of microcomputers and time-sharing systems led in the 1960s to studies into 'Man-Machine Integration' before further input from computing sciences, software engineering and information and systems sciences led HCI research towards personal computing applications and those singular problems. In particular:

- Computer graphics and the CRT (Cathode Ray Tube) and pen devices used in then current Computer Aided Design/Computer Assisted Manufacturing systems led to the study of interaction techniques in interactive graphics.
- Work on operating systems developed techniques for interfacing input/output devices, for tuning system response time to human interaction times, for multiprocessing, and for supporting windowing environments and animation.

In the early 1970s the minicomputer and then the microcomputer came into widespread use and a rapid expansion in computing power and functionality led to concerns about 'usability'. HCI has since been developing and applying design and evaluation methods to ensure that technologies are easy to learn and easy to use. A large body of evidence on what is 'good' and 'bad' usability was produced and methods for the production and analysis of such evidence were developed at that time.

1.4 Why it changed

Partly what changed was that the type of computer user and their expectations were no longer homogeneous. Originally, the users of computers were also the builders, engineers and software specialists, but as the minicomputer and remote terminal access to time-sharing mainframes brought computer usage out of the laboratory, scientists became superseded by commercial and business users and data processing professionals. John Carroll's essay on 'Where HCI came from'[4] also identifies the fortuitous coming together of a number of disparate strands in the late 1970s and early 1980s:

[4] *Carroll, J.M. (2009)* Human Computer Interaction (HCI). *Retrieved from Interaction-Design.org: http://www.interaction-design.org/encyclopedia/human_computer_interaction_hci.html*

- The emergence of personal computing, both that of personal software applications (text editors, spreadsheets and interactive computer games), and the development of robust computer platforms (operating systems, programming languages and hardware) made anyone with access into a potential computer user. This also 'vividly highlighted the deficiencies of computers with respect to usability for those who wanted to use computers as tools.'[5]
- Human factors engineering techniques for analysis of human-system interactions in command and control systems became viable in the wider context of user interactions.
- Developments in the methodologies and technologies of software engineering: on the one hand towards non-functional requirements which could include usability and maintainability; and on the other rapid development of interactive systems for CAD/CAM graphics and information systems.
- The newly formed discipline of cognitive science (incorporating cognitive psychology, artificial intelligence, linguistics, cognitive anthropology, and the philosophy of mind) provided the tools, skills and researchers to study cognitive engineering aspects of interacting with new technology.

[5] *Carroll, J.M. (2009) op. cit.*

> *All these threads of development in computer science pointed to the same conclusion: The way forward for computing entailed understanding and better empowering users.*[6]

[6] *Carroll, J.M. (2009) op. cit.*

In terms of technologies and innovative uses for computers, a number of pioneers were already, in the 1960s, laying the foundations of what would be the design science of HCI. All of these, and much of the earlier graphics work and activity, is discussed and illustrated comprehensively in HCI textbooks.

- At the Stanford Research Institute, Douglas Engelbart led a group developing the concept of augmenting human intellect via advanced computer tools. SRI pioneered the mouse and effectively invented WYSIWYG word-processing, multi-window display, and electronic meeting rooms.
- Ivan Sutherland's 'Sketchpad' system stimulated the development of computer graphics techniques before the technology was sufficiently powerful to develop them in full.

- Ted Nelson's focus was upon the way in which the computer could facilitate a new structural concept (hypertext) which overcame purely linear presentation of text.

1.5 HCI in the 1970s and 1980s

The starting decade for HCI as a separate discipline was the 1970s when leading research centres in two continents were set up: HUSAT in Loughborough, UK and PARC in California, USA. The HUSAT group played a leading role in applying the concerns, methods, and knowledge traditional to the field of ergonomics to the study of computer design and use. The Xerox Corporation's Palo Alto Research Center (PARC) built on Engelbart's work at the nearby Stanford University, and research there into networked workstations using the Smalltalk programming language led to the design of the Xerox STAR workstation, from which 'look-and-feel' the Apple Lisa and Macintosh were derived. Alan Kay and Adele Goldberg specified in outline a personal, portable information manipulator. The 'Dynabook' was the precursor of our portables, netbooks, laptops and e-readers. Elsewhere, other work was presented at small international workshops and the first input to European (German DIN) standards (for visual display terminals and for keyboard and keypad layouts) were proposed and the seeds of usability sown. Growth in personal computer technology led to an explosion in applications – text- and word-processing, graphics manipulation, the first interactive games, spreadsheets and the direct manipulation interaction paradigm.

The 1980s saw an increase in HCI publications and the first major attempt to formulate a theoretical basis for HCI (Card, Moran and Newell, 1983). The Model Human Processor[7] described human performance when interacting with computers as a model which could be used, together with operational models (Goals, Operators, Methods, Selection rules and Keystroke Level), to analyse human-computer tasks and to predict total task performance times. The human is seen as an information processor, with inputs (visual), mental processing, and outputs (keyboard strokes and mouse actions), which themselves 'input' information or data to the computer.

[7] *Card, S.K., T.P. Moran and A. Newell* The psychology of human-computer interaction*. (Lawrence Erlbaum Associates, 1983) [ISBN 0898598591].*

In the 1980s and 1990s the expansion of the communication networks linking computers together – that had started with the US Advanced Research Projects Agency's network, ARPANET, and continued with the world wide web from CERN – created new applications (electronic mail and computer conferencing) and the exponential growth of the internet we know today. HCI research concerns shifted towards communication between people enabled by computers; that is, how users might interact with each other via a computer. Researchers from social sciences (notably from Anthropology and Sociology) increasingly became involved and initiated a paradigm shift in studying how computers and interactive technologies were interpreted and appropriated by groups of users. An initial focus on Computer Support Co-operative Work was swiftly expanded to include ways in which to investigate user activities and to determine user requirements. The study techniques of social sciences were assimilated into the HCI mainstream.

This shift in emphasis to a social and emotional relationship with technology, matched the concerns of many in determining how to allow users to interact efficiently and effectively with a computer and saw the emergence of 'Usability

Engineering'. This approach was stimulated by the realisation that design, as a set of related practices in its own right, could provide valuable input to HCI research and to product design and evaluation. There was a move away from psychological and experimentally-based research and design, towards a quantitative but practical engineering approach to product design (i.e. early goal setting, prototyping and iterative evaluation). Together with a more commercial ethos (developing usable and functional products to improve productivity, integration of usability and product design teams and cost-benefit assessments of design decisions), usability methods began to develop beyond long-held academic roots towards the concepts of engineering and testing products and interfaces for usability. Usability testing had been focused on experimentation and inspections of interfaces based on good practice guidelines, but in the early 1990s new methods such as Heuristic Evaluation, cognitive walkthrough and usability questionnaires became widespread. Comparisons between usability evaluation methods and tools specific to a Usability Engineering approach were developed and a usability industry was created.

1.6 The situation today

We now view HCI design as looking past the simple interaction between user and computer/technology (and beyond even computer-mediated interaction between people) to involve wider social, cultural and aesthetic attitudes. Thus, by the mid-1990s designers and 'design practice' became the attitude of choice in HCI, emphasising practice-based approaches at the expense of information-processing models of user behaviour. HCI now encompasses many philosophies, perspectives and types of expertise. Different aspects of human–computer interaction mean using and learning different techniques, depending on different goals. Interaction Designers, HCI researchers and usability specialists now must be familiar with design, engagement, the practice of technology design ethnography and theories of social action, as well as with the more prosaic human and usability engineering techniques which have been progressively developed since the inception of HCI as a discipline.

Summary

The scope of HCI has progressed from ergonomics/human factors to human cognition and psychology; from effects on workflow and communication to effects on culture and society. We now investigate the relationships between people that computers and computer networks enable, such as patterns of behaviour between people and within social groups, and study the digital artefacts which shape aspects of our everyday lives.

A reminder of your learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- understand the definitions of HCI and ID and associated terminology
- demonstrate knowledge of the historical development of the discipline
- define and describe common HCI terms
- describe how future developments in interaction, design and technology link to the history of the study of ID and HCI.

Exercises

1. 'User interface design is all common sense!' Discuss.
2. Define the terms 'User Interface', 'Human Computer Interaction', 'Interaction Design' and 'User Experience'. Differentiate between them.
3. Investigate disasters or accidents that hinged on an HCI failure. What went wrong, and why?
4. Draw up a table showing the development of input devices over the last 30 years. What has been the major change? What might come next?
5. Informally assess the growth of mobile phone technology and applications in your home community. Who uses them, why, and to what purpose? How did those activities take place before the advent of mobile phones?

Sample examination question

What can the study of the historical roots of HCI bring to current design practices?

Further reading

For original articles and source material on the history of HCI

Baecker, R. and B. Buxton (eds) *Readings in human-computer interaction*. (Morgan Kaufmann, 1987).

Baecker, R., et al. (eds) *Readings in human-computer interaction*. (1995).

For general discussion on HCI and ID topics

Harper, R., et al. *Being human*. (2008) Appendix.

Lazar, J., et al. *Research methods in human computer interaction*. (2009) Chapter 1.

For basic concepts in user psychology and cognitive aspects of user interaction

Norman, K. *Cyberpsychology. Part ii: systems*. (2008).

Sutcliffe, A. *Multimedia and virtual reality*. (2003) Chapter 2.

Coverage of individual differences in psychology

Norman, K. *Cyberpsychology*. (2008) Chapter 9.

Oral history interviews from the Charles Babbage Institute, University of Minnesota

J.C.R. Licklider: www.cbi.umn.edu/oh/pdf.phtml?id=180

Allen Newell: www.cbi.umn.edu/oh/pdf.phtml?id=208

Ivan Sutherland: www.cbi.umn.edu/oh/display.phtml?id=100

Notes

To err is human; to forgive, design

Andrew Dillon

concept **design** development evaluation experience hci ideas interaction interface metaphors model process product prototypes requirements system techniques usability user work

Chapter 2
Interaction and design approaches

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- demonstrate a realistic appreciation of ID processes and activities
- discuss the variety of approaches to ID and the range of design techniques and methodologies
- describe how and when such methods are used in a design activity
- describe what a metaphor is, and identify its importance
- demonstrate practice in drawing, sketching and designing paper prototypes.

Essential reading

Note: Further reading and websites are listed at the end of the chapter.

HCI textbooks

Shneiderman, B., et al. *Designing the user interface*. (2009) Chapter 3.

Sharp, H., Y. Rogers and J. Preece *Interaction design*. (2007) Chapters 1 and 2.

Dix, A., et al. *Human-computer interaction*. (2003) Chapters 6 and 7.

Idea Generation

Buxton, B. *Sketching user experiences*. (2007).

Greenberg, S. et al. *Sketching user experiences: the workbook*. (2010)

Introduction

Design work in HCI and ID stems from a pre-existing design tradition and employs techniques used for many years in design disciplines which were not originally computer-based, such as architecture, urban design and planning, visual design, printing and graphics. The use of prototypes, storyboards, sketching, personas and even the words used to describe what Interaction Designers do (such as 'understanding the problem space' or 'generating ideas') derive from Design Studies, or from craft-based technical activities. More so than the wider discipline of HCI, the field of ID gives priority to designers and allocates them an essential role on every software and interface development team. It is about 'design' as an activity – providing designers with tools to operate effectively without compromising the overriding goals of making products which are usable, useful, and desirable. Interaction Designers – who can also be known as Usability Engineers, Visual Interface Designers, User Experience Engineers, or Information Architects – try to understand and shape human behaviour to achieve this goal, but the value of design, although now recognised as being an essential component of product development, is still often misunderstood. One way of looking at the nature of ID work is that it is a means of managing the complexity of interaction in an engineering-centric world; a way of connecting and thinking about people and their activities, emotions and experiences. It is about computing but also about language, communication and technology and the aesthetics of human interaction. For many in the ID world, it is the need to not lose sight of the elements of craft, execution, and appropriateness when working towards creating the 'great user experience'.

2.1 What is design?

All design is driven by requirements, the focus being on the core need – not on how it is to be implemented – since there may be multiple ways of achieving a goal. In order to design or build any type of system, two basic requirements exist: the designer must understand the fundamental and underlying requirements of the product, and then must 'develop' that product to fit those requirements as best as is possible. Such development includes creating or producing a range of representations (or 'models') of the system; some theoretical or conceptual ('mental models'), some illustrative such as storyboards or scenarios, and some physical, such as artefacts or prototypes. Effective representations should be accurate enough to reflect relevant features of the final system, but also simple and clear enough to avoid confusion. Such representations fulfil a variety of purposes, including:

- exploring the problem space
- communicating, illustrating and expressing ideas to others
- making predictions of user performance, effort and outcomes.

A design is a simplified representation of the desired outcome so one approach to developing effective representations is to ask a set of questions:

- What do you want to create?
- What are your assumptions?
- What are your claims for what it will do?
- Will it achieve what you hope it will? If so, how?

In thinking about and determining the answers to such questions, it is useful to have an overall strategy or plan to follow. In design disciplines this tends to follow a well-recognised sequence.

Look at similar artefacts » Analyse users' needs and abilities » Sketch different designs, make prototypes » Show to users and test » Build as a physical artefact

For simple design exercises and for small systems, this procedure is a tried and tested one and works well. However, with more complex systems and large-scale interactive applications, such a craft-based approach becomes time-consuming, unwieldy and expensive. A more systematic approach is required. Design of interactive systems then comprises the following activities or processes, the design of the final system being determined by the nature and content of the requirements definition. The overall process is, of course, similar to any software engineering or system development process and, later in this chapter, the relationship between SE principles and ID strategies will be explored.

An initial feasibility study » Followed by a requirements definition » The design is implemented » The implementation is tested » After the system has been in use for some time it is updated and maintained

In many situations, a feasibility study will not be relevant and will not be carried out, and for much of the design of interactive applications, maintenance is not a standard activity. This Study Guide concentrates on the intervening activities of 'Conceptual Design' followed by what could be termed 'Illustrative Design' before the final 'Physical Design' is produced.

Conceptual Design is essentially and simply about taking some requirements and turning them into a description of a future artefact (a system, device, application or interface) in terms of a set of integrated ideas and outline concepts about:

- what that artefact should do
- how that functionality can be achieved
- how the built artefact will behave or look – in a manner that will be comprehensible to its expected users.

This type of approach to requirements definition (which, in design communities, is not usually described in these terms) can be based on a set of design principles derived from an amalgamation of theory-based knowledge, experience and common-sense practice which can identify such specifics as:

- generalisable abstractions for thinking about different aspects of design
- the do's and don'ts of ID
- what to provide and what not to provide at the interface.

It can be shown simply as in Figure 2.1. By focusing on users and their needs, as stressed in the previous chapter, and specifying the usability and user experience goals that a design solution will achieve, four basic steps emerge within an iterative overall process of design refinement (i.e. repeatedly evaluating and correcting designs).

Material available only to students registered on this module.

[1] © Penn State University. Human Factors and HCI; http://www.personal.psu.edu/cwc5/blogs/coursedesign/media/110_s1406_g01.jpg

These four steps encompass many activities. Establishing requirements is about carrying out investigations to fill in any gaps in knowledge about the intended users, user characteristics, and the use of the product or the task. After settling on draft specifications (possibly with performance and cost estimates) an analysis of the practicality of alternative activities or design solutions can be carried out. This extensive conceptual design stage leads to the next stage of building and testing of prototypes as Illustrative Designs which can be tested or tried out and then modified and revised as necessary. Then testing – or evaluation – can be carried out on a succession of designs and repeated until a satisfactory result is achieved. At that point, the final design, or solution which equates to the Physical Design, can be constructed as an implemented artefact.

Determine user needs and establish requirements » Develop alternate designs that meet the requirements » Prototype, and evaluate with the different designs » Create and build a solution

In the remainder of this Study Guide specific chapters deal with each aspect of this activity. If the prime focus for HCI/ID design is to identify the user's needs and ensure that the designed functionality of the final built artefact matches that need, then Interaction Designers must develop a User-Centred Design mindset to think of the world in the user's terms and of what the benefits to users will be, rather than being technology-centred or feature driven. There are, of course, practical considerations to be taken into account in real-world design and many factors which affect design practices relate to issues such as cost, project size, development time and the design methodology to be used. However, the HCI stance, whilst acknowledging such needs, breaks down the design approach to two basic views and sub-activities embedded in a cyclical design process, as in Figure 2.2.

Understanding users and their tasks

- Task-centred system design process:
  - developing task examples
  - task scenarios and walkthroughs

Designing with the user

- User-Centred Design and prototyping:
  - User-Centred system design
  - low fidelity prototyping methods

- Evaluating interfaces with users:
  - observing people using systems via various methods
  - detecting inappropriate design and correcting by iterative design.

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[2] *© Penn State University. Human Factors and HCI; http://www.personal.psu.edu/cwc5/blogs/coursedesign/media/110_s1407_g01.gif*

In this chapter, User-Centred Design and prototyping concepts are highlighted and explained. In Chapter 3, the emphasis is on understanding users and their tasks. That chapter describes some techniques for ID requirements, essentially user requirements elicitation in a task-centred system design process. User Narratives (scenarios and personas) and Scenario-Based Usability Engineering are discussed in detail and prototypes in a more general fashion. Chapter 5 is concerned with the specific techniques for evaluating interfaces or designs with users in terms of usability and how well a design or implemented system meets its requirements. Usability itself is discussed in detail in Chapter 4.

Before going into more detail on UCD and prototyping, it is useful to take a step back and to look more closely at the process of design itself, namely, what is happening during that initial creative process and how to understand and describe it. The rest of the chapter will explain, first of all, some design principles which contribute to what can be called the ‘problem space’ or ‘design space’ and what some solutions may be. The various design methodologies commonly employed in HCI and ID will be introduced, before concentrating on the practicalities of the design activity itself and the range of techniques available for creating mock-ups, prototypes, storyboards (and similar) for testable Illustrative Design solutions.

2.2 Some principles of design

There are many designers and a multitude of approaches – extensively written-up – to ID and the creation of interactive computer applications. Many, however, are concerned solely with website design and web interactivity or with an area which has emerged called User Experience Design. Fewer concentrate on the wider issues of design for interaction as such, but useful information is given and many graphic design ‘tricks of the trade’ and generic guidelines can be found. There are too many books, guides, websites and blogs to discuss in detail in this Study Guide and it is impossible to advocate one over another: see the selected Further reading

[3] See also, http://www.rosenfeldmedia.com/zeitgeist/book/index

for this chapter and the web resources given in the Preface for listings.[3] It is strongly recommended that students study at least one of the suggested additional readings and access some design-oriented websites to gain a rounded view of this area.

In terms of an overall design approach, we will concentrate on a design basis put forward by Donald Norman, an early HCI advocate, guru and proselytiser. The focus is on a core set of principles, as discussed in his series of books on the design of everyday and computer artefacts and on his discursive website.[4] Numerous amendments have been made to such principles over the years but these basic elements will help to clarify later expansions. The concept of 'affordance' is an especially important one to grasp.

[4] http://www.jnd.org/index.html

Consistency

'Consistency' is one of the most widely known user interface design principles. Simply put, it means that mechanisms should be used in the same way wherever they occur. Consistency is expressed as:

- Internal consistency of a design with itself.
    - For example, the use of consistent command names, selection of menu options, size and style of icons, number of mouse clicks to select items, etc.
- External consistency with other interface designs and other systems.
    - This allows users to switch between systems and applications without undue effort.
- External correspondence of a design to features in the world beyond the computer domain.
    - Examples include the desktop metaphor and the arrangement of arrows on cursor control keys to match the arrangement of compass point arrows on a map.

Trade-offs have to be made between internal and external consistency but what is more important is the context of use (i.e. users' tasks and the way they are best carried out). This may mean that aspects of interfaces are actually internally inconsistent but consistent with the way users perform their tasks.

Feedback

'Feedback' can be defined as '...sending back to the user information about what action has actually been done {and} what result has been accomplished'.[5] Another way of describing it is in terms of its absence, for example, using a cash machine without a screen would be virtually impossible as there would be no feedback as to which request had been accepted or if any mistakes had been made. Feedback is an integral part of the interface. A key aspect of designing metaphors for both everyday applications and novel interfaces is to provide appropriate visual, auditory and tactile feedback. Examples of good immediate feedback at the interface include simple ones like those listed below:

[5] *Norman, D. (2002).* The Design of Everyday Things. *(Basic Books).*

- displaying the details of the date and time when a file was updated
- the percentage bar showing the rate and progress of a process that is being performed

- 'Busy' icons showing that the system is processing and temporarily unable to respond to other commands
- an effective sound tone which acts as a warning and attracts a user's attention
- the clicks of simulated key presses on touch screens.

More examples can be found in all of the recommended HCI textbooks and in Norman's writings and website. A pertinent and timely discussion of the new ways in which users can interact with the latest mobile phones and the Apple iPad highlights just these issues and why they are so important.[6]

[6] *Norman, D. and J. Nielsen (2010), 'Gestural Interfaces: a step backwards in usability'; http://www.jnd.org/dn.mss/gestural_interfaces_a_step_backwards_in_usability_6.html*

Mappings

'Mappings' refer to the spatial and conceptual relations between different parts of a system or between controls and their outcomes. Mappings are 'good' if they appear natural and intuitive to users. Poor mappings are when the relations are inconsistent or incompatible – Norman uses the example of light switches, lift controls, door handles and car indicators and their relative directionality to illustrate this point.

Affordances

For controls to be easy to use, mappings should be obvious and demonstrate a good 'affordance'. The term originally referred in psychology to actionable properties between the world and an actor – as a perceived relationship between them. Norman first linked the concept to design stressing that what he identified was actually 'perceived affordance'.

> *'I introduced the term affordance to design … the concept has caught on, but not always with true understanding. Part of the blame lies with me: I should have used the term "perceived affordance," for in design, we care much more about what the user perceives than what is actually true. What the designer cares about is whether the user perceives that some action is possible (or in the case of perceived non-affordances, not possible). In product design, where one deals with real, physical objects, there can be both real and perceived affordances, and the two need not be the same. In graphical, screen-based interfaces, all that the designer has available is control over perceived affordances. The computer system, with its keyboard, display screen, pointing device (e.g. mouse) and selection buttons (e.g. mouse buttons) affords pointing, touching, looking, and clicking on every pixel of the display screen. Most of this affordance is of no value. Thus, if the display does not have a touch-sensitive screen, the screen still affords touching, but it has no result on the computer system. … All screens afford touching: only some detect the touch and are capable of responding. But the affordance of touchability is the same in all cases. Touch sensitive screens often make their affordance visibly perceivable by displaying a cursor under the pointing spot. The cursor is not an affordance; it is visual feedback. In similar vein, because I can click anytime I want, it is wrong to argue whether a graphical object on the screen "affords clicking." It does. The real question is about the perceived affordance: Does the user perceive that clicking on that location is a meaningful, useful action to perform?'*[7]

[7] *Norman, D. http://www.jnd.org/dn.mss/affordances_and.html*

For example, does the handle on a door indicate if it should be pushed or pulled, or does a slider or scroll bar intuitively indicate its direction of movement? If not, does it require extra perceptual cues (such as the addition of direction arrows) or instructions to make its use transparent? The definition which has come to be used is: 'the behaviour of an object is that which is permitted and perceptually obvious'. The term refers to the properties of objects – what sorts of operations and manipulations can be done to that object. What is important is the 'perceived affordance' (the functionality that is suggested by an object's placement, or its look, or its associations, or the feedback it provides).

Constraints

When designing an interface it is important to think about how the information to be displayed or presented is constrained so that it exhibits good affordance and can be used easily and simply. Constraints limit the number of possibilities of what can be done with an object. Norman originally suggested four main types:

Physical, Semantic, Cultural and Logical.

- Physical Constraints restrict the possible operations of an object:
    - A key can only fit a lock in a certain way.
    - The scroll bar in graphical interfaces is physically constrained by the shaft that restricts movement of the scroll box to one of two alternatives; up or down (vertical bars) or left or right (horizontal bars).
- Semantic Constraints depend on the semantics (the 'meaningfulness') of the situations which users know about. They depend upon a user's knowledge of the world and on how the computer system is being used.
    - The icon of a wastebasket is semantically constrained because it is represented upright on the desk top. It could be represented on its side but this would contravene our knowledge of how waste baskets are used.
- Cultural Constraints consist of information and rules that help us to know what to do in social settings. Many icons have been designed to capitalise on cultural conventions.
    - Pictorial signs on toilet doors signal that one set is meant for women and the other for men.
    - The Apple Macintosh had a warning icon that consists of a bold outlined icon with an exclamation mark.
- Logical Constraints work through constraining the order, position or location of objects.
    - The order in which items are displayed in menus is logically constrained by appearing as lists of horizontal items. It would seem illogical if the items were randomly displayed across the screen in any direction.

2.3 Problem space and design space

An important issue in design is to identify and analyse what is called the 'problem space' and to move from that to the 'design space'. In a similar fashion to that of 'identifying the design artefact' and creating representations of it, questions can be asked of a UX design:

- Are there problems with an existing product or user experience?
- Why do you think there are problems?
- How do you think your proposed design ideas might overcome these?
- When designing for a new user experience, how will the proposed design extend or change current ways of doing things?

Having a good understanding of the problem space, that is specifying what is being done, why, and how it will support users in the way intended, can help inform the design space. Before deciding upon what kind of interface, behaviour and functionality to provide, it is important to develop a conceptual model and to think about how the system will appear to users. A conceptual model is not a description of the user interface but is a high-level description of a product showing what users can do with it and the concepts they need to understand how to interact with it. A conceptual model is the 'mental model' users have of the 'system' and how it works. It is built up from actual interaction with the system through what is termed the 'system image' (i.e. the interface and many other aspects such as physical properties, documentation, icons and instructions). The designer's mental model is how user interaction with a system is envisaged. Actual realisation of this concept is what is produced as the 'system image' – an interpretation of an idealised interface. This concept is discussed widely in HCI textbooks and is often illustrated as in Figure 2.3.

Material available only to students registered on this module.

[8] *© interactiondesign blog.com, 2008–09. How designers communicate with users; http://www.interactiondesignblog.com/2008/06/how-designers-communicate-with-users/*

Decisions about the conceptual design must be made before commencing any Physical Design and, if based on the approach outlined above, will be effective:

> *The most important design tool is that of coherence and understandability which comes through an explicit, perceivable conceptual model. Affordances specify the range of possible activities, but affordances are of little use if they are not visible to the users. Hence, the art of the designer is to ensure that the desired, relevant actions are readily perceivable.*[9]

[9] *Norman, D. http://jnd.org/dn.mss/affordance_conventions_and_design_part_2.html*

2.3.1 Metaphors

'Interface metaphors' and analogies are commonly used as part of a conceptual model to convey to users how to understand what a product is for and how to use it for an activity. They are concepts that users are exposed to through the product, the relationships between the concepts and the mappings between the concepts and the user experience that the product is designed to support.

Based on Norman's constraints, an interface metaphor is designed to be similar to a physical entity but to also have its own properties. Metaphors exploit a user's familiar knowledge, helping them to understand 'unfamiliar' functionality since people find it easier to learn and talk about what they are doing at the computer interface in terms familiar to them. The general benefits are that metaphors:

- help users understand the underlying conceptual model
- make learning new systems easier
- are accessible to a greater diversity and number of users.

Metaphors can be powerful design tools, able to generate new associations from existing knowledge but the ability to communicate complex ideas can also have the effect of constraining thought and the user's own discernment. Sometimes connections that may make sense in the source concept do not match up to the target concept. The use of metaphors in HCI and ID has been extremely successful in some cases – the desktop metaphor is now endemic – and a distinct failure in others – the Microsoft Office paperclip ('Clippy') is now universally derided. A number of texts listed as Further reading give guidance on how and when to use interaction metaphors and how to design with metaphors.

Problems with interface or interaction metaphors counter some of Norman's guidance in that they can:

- constrain designers in the way in which they conceptualise a problem space
- limit designers' imagination in coming up with new conceptual models
- conflict with other design principles
- break conventional and cultural rules
- force users to only understand the system in terms of the metaphor.

Metaphors can go wrong in so many ways – or can simply become mixed and confused – so it is crucial that the choice of metaphor should start with the expectations, needs, desires and actions of those involved. One author of a current Ubiquitous Computing and UX design book[10] highlights a

[10] *Kuniavsky, M. (2010).* Smart Things: Ubiquitous Computing User Experience Design.

way of thinking about metaphors as a design tool by asking the following practical questions:

- What is the comparison that this metaphor is making? What class does it say that the design and the metaphor belong to?
- What is the list of tools and activities associated with the source concept? How would those map to the experience being designed?
- What are the visual images the source concept evokes? What are the interaction patterns that it implies? Are there necessarily positive outcomes to those patterns? Negative ones?
- What is the purpose of using the metaphor? What exactly do you need it to accomplish? What associations is it supposed to evoke and what actions will the metaphorical associations make easier?
- What are the boundaries of the metaphor? At what point do the differences between domains become so great that the metaphor hurts more than it helps?

2.4 Design methodologies and approaches

As discussed in the previous chapter, we saw that HCI was derived from software sciences combined with cognitive sciences. The first design methodologies generated were based upon classic software engineering models – partly because of the type of computers in use and partly because the early interfaces were to CAD and graphics applications and to control systems which required an understanding of how data was input and the type of input devices used. As software became more of a design science and issues of practical management and efficiency became more prevalent, newer design methodologies were developed, supported for the most part by large corporations who were in the business of developing and selling first mini-, and then microcomputers. Mass production and popularisation of personal computers and a change in the styles of computer companies from large mainframe retailers to small start-ups (such as Microsoft and Apple once were) meant that hardware development inexorably led to newer forms of programming languages, different types of operating system and newer applications (such as spreadsheets and drawing packages like MacPaint) with a wealth of new input devices and interface display options. This heralded a gradual change in the focus of software engineering, away from programme efficiency, code optimisation and even 'elegance' to the user, usability and, ultimately, the user experience. At that stage, creative designers began to get interested in something more tangible and to use the devices themselves.

HCI in the 1980s essentially grew out of graphics research and applications whilst Usability Engineering was more of an outgrowth from the impacts of increased user input, from developments in requirements capture methodologies and from standardisation and the use of usability metrics. Pre-existing standard methodologies and frameworks were subjected to user-centred modifications to enable newly-created 'discount' usability evaluation methods to be used for obtaining structured feedback from users about aspects such as usability and accessibility. This required that a cycle of prototype building and testing be done iteratively. With each iteration, prototypes move from low-fidelity sketches or simulations to more functional versions of a final design. The rationale is that the transition from formative evaluation of design concepts to summative evaluation of a final artefact minimises the emergence of unexpected usability problems or unforeseen user responses.

Figure 2.4 illustrates the iterative development cycle.

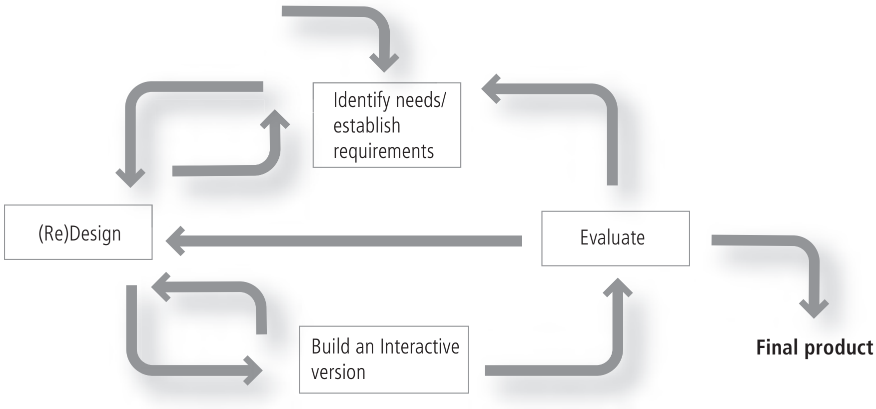

Figure 2.4 Iterative Design Model

A paradigm change occurred with the move towards concepts of experience-centred design. The inclusion of techniques such as ethnography – from sociological and anthropological disciplines – and technological developments led to a more co-operative design focus. ID and UX design came to the fore as the conceptual and methodological contributions of the 1990s addressed the challenge of understanding the user and the context of use. It became evident that there was a need to study work in detail and 'in situ', determining what people actually 'do' in their work practices before making design interventions. Such approaches were part of a widespread movement to assist and encourage researchers 'in the field' to communicate their findings to systems designers and software developers, and to help those designers and developers to represent users and the use context during the design process so that user-centred decisions could be made. These representations include personas and Scenario-Based Design and are discussed in the next chapter. To further incorporate users into the design activity and to foster greater participation meant input from other design disciplines and so Participatory Design – from Scandinavian work practices – became a greater feature in ID.

2.4.1 Expansions of software engineering models

ID can make use of a number of different software life-cycle models, the simplest being that described in general terms in Section 2.1, but with the addition of an iterative cycle to the process.

ITERATE: Needs identification » Design »

Build alternative prototypes » Evaluate

The Waterfall and the Rapid Applications Development models, derived from 1970s software engineering, enforced linear progression from requirements to testing with documented completion criteria at each stage. Although there is the possibility of some user involvement at the requirements stage, the design is forced into smaller project 'boxes' and cannot be easily iterated, as shown in the step process below.

Requirements » Design » Detailed Design » Code » Integration » Implementation » Test

A need for design approaches which could cope with iteration and complexity was thus recognised and 1980s HCI developers started to build upon pre-existing SE developments. The Spiral and the Star Models were accepted as suitable approaches for UCD with its emphasis on determining users' needs, understanding the user and task context and designing products from that basis rather than from developers' preconceptions or rigid procurement briefs. Both are extensively described in the recommended textbooks.

Spiral Model

One model, developed by Hartson and Boehm-Davis[11] and shown in Figure 2.5, is in that spiral shape to reduce the risks of early design commitment. It is derived from an earlier model[12] in classic waterfall software development lifecycle methods and describes an iterative, incremental approach to dealing with change and managing risk. It is a sequential but iterative approach since incremental development allows developers to define and implement features in order of decreasing priority. An initial version of the system is developed, and then repetitively modified based on input received from external (customer or user) evaluations. The development of each version of the system is carefully designed using the steps involved in the Waterfall Model. With each iteration around the spiral (beginning at the centre and working outwards), progressively more complete versions of the system are built. Thus the model relies heavily on prototyping in that it explicitly encourages alternatives to be considered.

[11] *Hartson, H.R. and D. Boehm-Davis 'User interface development processes and methodologies',* Behaviour & Information Technology*, Volume 12, Issue 2, March 1993, pp.98–114.*

[12] *Boehm, B. 'A Spiral Model of Software Development and Enhancement'* IEEE Computer*, 12(5), May, 1988, pp.61–72.*

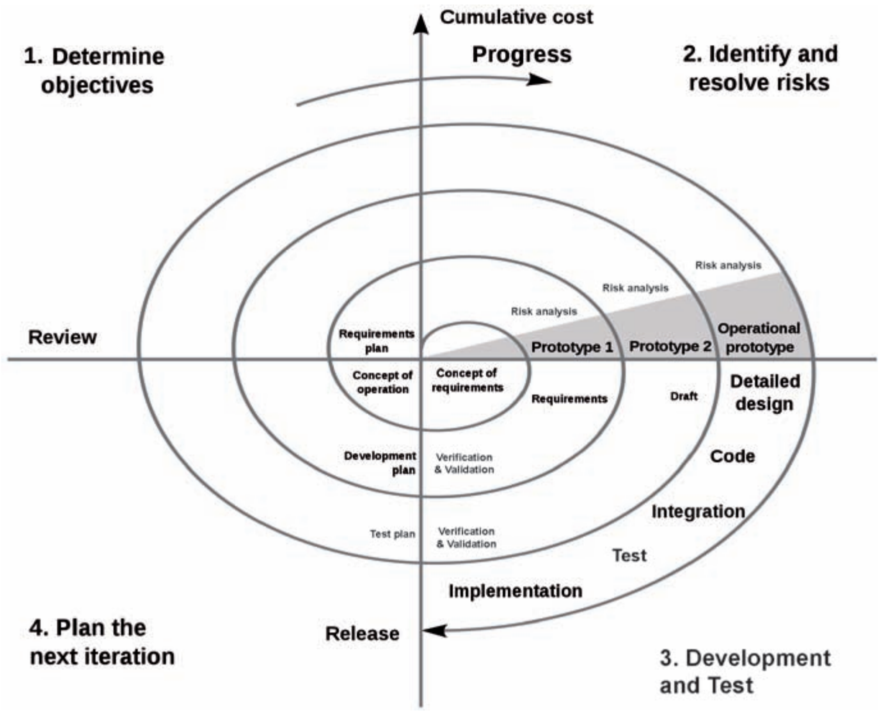

Figure 2.5 Spiral Model [13]

[13] *Source: http://commons.wikimedia.org/wiki/Category: Spiral_model_of_Boehm/ Reproduced under a Creative Commons Licence.*

Star Model

Another model, more useful to HCI design practices, is one that constantly returns to evaluation as its core process: the Star Model from Rex Hartson and Debbie Hix, shown in Figure 2.6. It was derived from empirical studies of how interface designers actually worked. The fundamental concept is that the design of interactive systems typically does not follow a specific order of steps but can start at any stage, always followed by an evaluation activity. Evaluation is the central phase in the development cycle. Development can start from any point in the 'star' and any stage can be followed by any other stage, but evaluation is always carried out before moving to a new stage. The requirements, design and product gradually evolve, becoming increasingly well-defined. This cyclic design emphasises rapid prototyping embedded with alternating waves of analytic (top-down) and synthetic (bottom-up) design steps. The Star Model can give the user a significant role throughout the project since evaluation is so central to the cycle and, since evaluation can be based on any representation of the system, this model too relies heavily on prototyping.

Material available only to students registered on this module.

[14] *Hix, D. and H.R. Hartson.* Developing user interfaces: Ensuring usability through product and process*. (John Wiley and Sons, 1993).*

These two models provide very specific ways of managing the design process and are intended to maximise User-Centred Design strategies. More complex software engineering methodologies are constantly being developed and refined but this Study Guide does not cover any other SE-derived approaches or techniques. Interested students can find further detail in the recommended textbooks and in the Further reading list.

2.4.2 User-Centred Design

User-Centred Design or User-Centred System Design developed from the belief that, in order to design truly usable systems, designers and developers needed to have a much clearer understanding of what the users of a proposed system or application actually wanted from it. Together with considering seriously whether those users would be able to understand and easily use the proposed design, and a better appreciation of how work tasks were currently carried out, this approach grounds the design process in information about the people who will use the product. A research impetus to make digital technologies accessible to, and usable by, ordinary people created a design focus on users throughout the planning, design and development of a product. UCD came to prominence through the design and development of the Olympic Messaging System[15] used at the 1984 Olympics in Los Angeles.

[15] *Gould, J.* et al. *'The 1984 Olympic Message System: a test of behavioural principles of system design'* Communications of the ACM *Vol. 30 (9), September 1987); www.research.ibm.com/compsci/spotlight/hci/p758-gould.pdf*

It identifies the needs of users and those impacted by the introduction of new systems or applications at an early stage and involves a task and a requirements analysis. It is an inherently iterative process emphasising early testing and evaluation with real users. UCD focuses specifically on making products usable by involving users to obtain feedback through the use of prototypes. As with other approaches, designs are modified in light of the user feedback, and the process aims to foster better communication. It has become the classic HCI approach to interactive design, concerned with the specification, delivery and evaluation of systems as a whole. It is the basis for Usability Engineering and for usability and HCI standards (as discussed in Chapter 4). The way in which user requirements can be elicited, and how user needs can be determined, is covered in the next chapter and evaluation processes are detailed in Chapter 5.

Allowing users to become involved in the design process means that designers must:

- Focus early in the design process on users and their tasks.
- Identify appropriate interface functionality to increase user satisfaction.
- Decide how best to utilise design decisions to produce systems which are acceptable to users.
- Establish usability criteria and responsibilities before irrevocable hardware and software decisions are made.
- Measure, record and analyse users' reactions and performance to scenarios, simulations and prototypes.
- Design iteratively – when problems are found, fix them and carry on testing.

Including users in the whole design process, either as co-designer or through evaluation of prototypes can be difficult to achieve. Coordinating high levels of real user involvement involves many practical problems in terms of time, money, scheduling, and, critically, consistency of user input throughout the project. One solution is to use techniques which bridge designer and user cultures. One such is Participatory (or Participative or Collaborative) Design.

2.4.3 Participatory Design

The Scandinavian tradition of social democracy enables the democratisation of computerisation by involving workers and trade unions directly in the design process. It is based on an action research model in which researchers and workers collaborated to produce improved conditions for the workers/users through active and continued consultation and collaboration between designers/developers, managers and workers. This type of approach emphasised workers' own experiences and required that researchers get involved with and develop a commitment to them in order to understand and change their work experience and conditions. The Participatory Design technique enables effective communication between two populations (workers/users and designers) about proposed designs which impact on users' working lives. Social and organisational aspects are emphasised but the process is based on studying designs, model-building and analysing new and potential future systems. Domain experts are part of the design and development team throughout the process. User involvement is built into an overall team process, the team being made up of a range of participants representing the whole user community assisted by a facilitator. The practice of integrating usability into the Participatory Design process has been evident since the early 1980s and one approach to usability testing is to have users engage in co-operative evaluation of prototypes. Successful projects, where information about users was integrated well into the design showed dramatic improvements in the resulting design, with increased user satisfaction and better business results. The benefits of such user/worker involvement can be seen by reference to the high costs of bad design:

- heightened visibility through electronic monitoring
- displacement
- intensification
- reduction or redefinition of skill
- loss of work autonomy and control.

The creation of shared representations in a process of 'co-design' is by using simple tools (such as scenarios or envisionments based on paper or video mock-ups) to explore innovative solutions. The use of personas, user profiles and scenarios derived from a contextual inquiry (essentially, abstract representations of users to guide design, which is discussed in detail in Chapter 3) and newer co-operative design techniques has been growing in recent times.[16] Focus groups, focus troupes and the use of video prototypes means that users are presented with a theatre play or a video of a scenario and discuss it in detail after the performance. Extensions to this approach are to let users develop video scenarios by themselves: such generative tools[17] incorporated into participatory development of scenarios and prototypes help engage participants and enhance the shared context.

[16] *Grudin, J. and J. Pruitt 'Personas, Participatory Design and Product Development: An Infrastructure for Engagement',* Proceedings of Participatory Design Conference *Malmo, Sweden. (ACM Press, 2002).*

[17] *Yylirisku, S., K. Vaajakallio and J. Burr (2007) 'Framing innovation in co-design sessions with everyday people'; http://www.nordes.org/data/uploads/papers/104.pdf*

Attributes of Participatory Design

- The goal of the activity is to improve the work life of users.
- The activity is collaborative, with all goals and decisions actively negotiated and not imposed.
- The activity is iterative – ideas are tested and adjusted by seeing how they play out in practice.

It can be a controversial activity, with both benefits and drawbacks. More user involvement brings:

- More accurate information about tasks
- More opportunity for users to influence design decisions
- A sense of participation that builds users' investment in successful implementation
- Potential for increased user acceptance of the final system.

On the other hand, extensive user involvement may:

- Build opposition to implementation
- Lengthen the implementation period
- Be more costly
- Show that organisational politics and preferences of certain individuals are more important than technical issues
- Build antagonism with people not involved or whose suggestions are rejected

- Force designers to compromise their design to satisfy incompetent participants
- Exacerbate personality conflicts between design-team members and users.

2.4.4 User Experience Design

Experience-based design (see Figure 2.7) involves both the observation of, and involvement in, users' everyday working life (these techniques are described in Chapter 3) in order to be able to represent what users' activity and response to the systems they use actually is. Designing in an experience-centred way means that '*the view of the human in HCI becomes richer and more open once this point of view is adopted, and it thus offers greater surplus and a richer potential with which to work as designers.*'[18] As these authors identify, key landmarks are in:

- *Valuing the whole person behind 'the user'.*
- *Focusing on how people make sense of their experiences.*
- *Seeing the designer and user as co-producers of experience.*
- *Seeing the person as part of a network of social (self-other) relationships through which experience is co-constructed.*
- *Seeing the person as a concerned agent, imagining possibilities, making creative choices, and acting.*

A number of influential authors[19] have described, with reference to how designers actually create designs, how interface development can be seen as an issue in design, claiming that multidisciplinary teams deal best with designing interfaces to systems with a high level of information content or complex web sites, games, and data visualisations. This work created the idea of ID as bridging the boundary between software interface design and media design. Others, more recently,[20] have argued that:

> '*Designing for usability, which had been the primary objective of HCI research and practice, was only one of the many values that User-Centred Design could focus on. Designing for fun, enchantment, adventure, and excitement were equally valid self-centred goals that were resonant with the shift of emphasis from designing office- and work-oriented products to designing for home-based, leisure, and entertainment products.*'

Norman, in his book[21] – with the slogan 'beautiful things work better' – opened up an interdisciplinary debate around beauty and pleasure as a design value. Investigating and explaining the relationship between aesthetics and usability has spread throughout the design research community and feeds directly into the way in which the design of future products (as described in Chapter 9) can be carried out. As one of the major proponents[22] has written:

> '*It takes an experiential approach, putting experience before functionality and leaving behind oversimplified calls for ease, efficiency, and automation or shallow beautification. Instead, it explores what really matters to humans and what it needs to make technology more meaningful.*'

[18] *Wright, P. and J. McCarthy* Experience-centred design: designers, users, and communities in dialogue. *(Morgan and Claypool, 2009) [ISBN 9781608450442].*

[19] *Winograd, T. et al.* Bringing design to software. *(Addison-Wesley, 1996) [ISBN 9780201854916].*

[20] *Blythe, M. et al.* Funology: from usability to enjoyment. *(Springer, 2003) [ISBN 9781402012525].*

[21] *Norman, D.* Emotional design: why we love (or hate) everyday things. *(Basic Books, 2004) [ISBN 9780465051366].*

[22] *Hassenzahl, M. (2010).* Experience design: technology for all the right reasons; *http://hassenzahl.wordpress.com/*

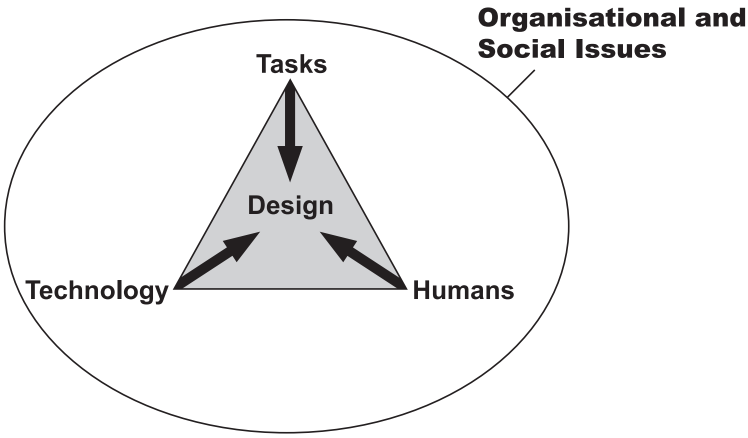

Figure 2.7 User Experience Design

2.5 The design activity

Alongside making requirements identification as real as possible for designers is the need to make emerging design ideas as concrete as possible for potential users, allowing them, as has been stressed before, to be consulted during the design process. A fundamental part of an iterative design process is to create an early partial implementation of the artefact or system: a prototype. Rapid prototyping techniques allow prototype development early in the design process (before time and money have been spent making irrevocable design decisions) with the effect that design solutions can be easily and quickly shown to users at an early stage. Rather than build a fully operational system, it is faster to prototype first, and test out that design solution (first with the design team and then with users for their reaction), since people are generally good at reacting to concrete designs and physical mock-ups or partial prototypes. In addition to fostering teamwork and communication by involving users, this process means that redesign based on feedback occurs until an acceptable state is reached. Of course, a newly generated design may be worse or may have consequences but this is part of the skill set of the Interaction Designer in choosing what to prototype, based on knowledge and experience. With prototyping and testing activity, it is obviously crucial to keep track of the ‘Design Rationale’ (i.e. the reasons for design decisions and changes, why those changes were made and what the implications are). This Study Guide will not discuss Design Rationale or its associated research areas but interested students are referred to the Further reading list or an HCI Glossary.

2.5.1 Prototyping

Prototypes are not just ‘cut-down’ versions of a finished artefact. They can be anything from a cardboard mock-up or a cartoon-like storyboard to a software presentation that simulates a system’s response to user actions. For user interfaces, representations of the artefacts to be designed can be built with different media, from simple paper to sophisticated interactive screen-based simulations. Mock-ups can range from concept demonstrators with limited pre-scripted functionality to partially implemented software capable of demonstrating interaction sequences and user experiences. The differences between the techniques are in the media employed, production costs, and the realism or fidelity of the representation of the proposed design. A classification and discussion of types of prototyping technique is covered in Section 3.7 of the next chapter but, for now, it is important to remember that all such representations result from the creative design process and are intended to illustrate concrete aspects of a design. Some variants are given below.

- Text descriptions of tasks
- Flow diagrams or outlines showing task structures
- Screen sketches
- Storyboards
- Paper prototypes
- Wireframe models – to make mock-ups
- Screen-based animations
- Walkthroughs
- Rapid video prototypes
- Virtual/Augmented/Mixed Reality mock-ups
- Wizard-of-Oz simulations
- Executable prototypes

Starting out

This section will discuss only a few of these in detail; some are self-explanatory whilst the specific technique of Wizard-of-Oz simulation is covered in Section 5.2.1. The recommended textbooks all cover prototyping and design realisation in extensive detail.

The power of a prototype is that it can create a focus for design conversations based upon something that is tangible and visible; it is a concrete example which provokes a user reaction. The process ideally starts with creative brainstorming to map out a space of ideas, concepts and potential designs (discussed in Section 2.6 in more detail). Once such design conceptions have been assessed and prioritised by a design team, the process of design realisation involving a user–designer dialogue can begin. Scenarios and personae (see Section 3.6) can be collected together in order to provide material for stimulating further discussion and for instantiating concepts which can be tested and iterated. Such design realisations are created in iterative cycles using scenarios and personae and, sometimes, examples of other good designs for inspiration. Initial ideas lead to design exploration in which storyboards, wireframes or mock-ups can be demonstrated to users in interview or workshop settings to elicit their feedback and to allow them to contribute ideas and participate personally in the design process. Representations of designs become more detailed and sophisticated as cycles of creation and evaluation occur; storyboards and early mock-ups are refined into software prototypes. Prototypes tend to be evaluated through a more formal test process (see Chapter 5) of evaluating usability.

One choice which may have to be made is the extent to which the process should be user-centred or designer-led. We advocate User-Centred Design but it can produce quite conservative solutions since users may fix their requirements and visions on prior experience rather than exploring new directions. On the other hand, designer-led exploration can develop rather too complete and complex prototypes based on idealised designs or novelty for its own sake. It is generally better to have design-led exploration when products are being developed for new markets whilst a user-centred approach may be best for applications tailored to specific users and tasks.

Mock-ups

Simple mock-ups can be very useful in allowing users and designers to interact directly, to engage in a conversation and to exchange views. Simple mock-ups can be easily constructed from common office material, since the demonstration focus is on the flow of interaction rather than the visual look and feel of an interface. A more complex physical model is an object which users and designers can interpret in various ways according to their interests and viewpoints: different people may take different meanings from the same model and highlighting such differences aids understanding. Physically interacting with mock-ups means that the visual, auditory and tactile senses are all used, more so than with a paper or screen-based design. As with all types of prototyping, this can help elicit feedback that becomes more and more focused and can strongly encourage creativity.

Scenarios and storyboards

Scenarios and personae are inputs to stimulate design exploration; storyboards are the initial outcomes of design (they may also be called design sketches or wireframe models). Scenarios are mini-scripts that describe situations where a product is used; storyboards visualise a narrative. These techniques can work together to simulate key moments in application usage and to communicate key aspects both within the design team and to those external to it.

Scenarios can take almost any format that 'tells a story' about a product or application and people or users. A scenario will involve a range of parameters, always with people or characters standing for the user and often with objects, circumstances and time as the additional elements. Scenarios should be detailed, evocative and focused on the value of the experience to users. Multiple scenarios can be envisioned; the same people can experience different situations, and the same situation can be played out with different products or users.

> '*For one single product, a team can write scenarios that describe, say, an ideal everyday use situation, a first use scenario and an edge-case (someone who uses the product ten times as heavily as an everyday user, for example). The point is to tell detailed stories about a specific user experience and to use the process of creating the scenario as an opportunity to better understand the details of the experience and to share that understanding among the team.*'[23]

[23] *Sutcliffe, A. (2009).* Designing for User Engagement: Aesthetic and Attractive User Interfaces.

The storyboard concept itself originates in animation and cartooning – ideas for a storyline or script were sketched as a set of drawings. In ID, storyboards usually illustrate 'snapshots' of interaction which is related to the users' tasks or what the application does, or a script of how the product will be used. Since many user experiences can have multiple outcomes based on the users' choices, multiple storyboards can describe possible experiences, or interactions.

Storyboards can be hand-drawn or created as PowerPoint animations or videos which are 'walked through' in order to elicit user feedback and to focus discussion on critiquing and elaborating the design. Such screen-based presentations (and many paper-based ones) can include clip art, found photographs, staged photographs, computer animations, dialogue clips and even cartoons. More recently, video mock-ups and storyboards are being used, since film and video has become so much easier to create on modern computers. They are all useful in showing multiple possible paths through an interaction, interactively, in a presentation that demonstrates how different decisions lead to different experiences. However, a limitation of storyboarding in this manner is that complex interactions can be hard to instantiate since interaction can only be represented by scripted animations or explained using paper materials. Complex ideas which relate to user immersion, presence or flow (see Section 9.2) are often very important parts of interactive design but can be extremely difficult to illustrate with sketches and screen-based presentations. Storyboards do, however, allow the rapid iterative exploration of design ideas we have identified as being a critical part of the design process, and they can be readily transformed into software-based prototypes which allow more interactive functionality to be demonstrated.

Simulations

> '*Simulation is a kind of drama. By using media to create a sensation of realness, an audience can suspend disbelief long enough to provide authentic reactions to the unfamiliar things they're seeing and hearing. Simulation takes place in the world at large, not just on a screen. In fact, simulating Ubiquitous Computing User Experience Design has much in common with theatre. In this theatre of design, devices are props, environments are stages, users are actors and user experiences have internal narratives.*'[24]

[24] *Sutcliffe, A. (2009).* Designing for User Engagement: Aesthetic and Attractive User Interfaces.

Newer techniques that are more appropriate and suitable for Ubiquitous Computing (see Chapter 9) and for ID and UX design make use of concepts appropriated from drama, theatre, film and role-playing games. A short play is produced acting out in a more realistic setting the user interaction scenario. The viewing and/or the production of the play tends to focus discussion as users evaluate ideas, but bring in their own context to the discussion too, producing new insights and ideas. There are many variants: professional actors (see Section 6.2.2) may be chosen over letting users act out parts for themselves. In that case there may be choices:

- give out a role or persona
- allow wholly improvised play (usually to foster evaluation and idea generation)
- allow participants to develop their own scenarios
- 'freeze' the action, discuss and change events, and then replay
- add further objects to the scene (changing the product ecology).

Such ways of prototyping interaction are constantly evolving as technology itself develops and different interaction styles become more common. These techniques can also have other uses – in more specific product evaluation and for demonstrating user difficulties by highlighting specific and problematic situations and settings (see Chapters 6 and 7).

2.6 Idea generation

The design process begins with generating and producing ideas for potential designs and this activity, too, can be achieved in numerous ways. There are as many inspirational brainstorming techniques as there are potential designs, and the recommended books chosen for this part of the course are those by Bill Buxton,[25] a polymath HCI, graphics and design pioneer. His first book describes how to think about the role of design and the principles of 'sketching'. He discusses some of the techniques mentioned above for prototyping and envisioning user experience in an entertaining and informative fashion. The associated workbook contains practical examples, suggestions and guidance on how to generate ideas. Buxton explains his views on sketching in more detail and identifies various methods for capturing aspects of the real world and possible user interactions, from single images to interactive modelling for prototyping. The concluding part highlights and describes some techniques of storytelling, critiquing and evaluating which are used to share the ideas generated with others, whether they be users or the design team. Together they are the best sourcebooks currently available and the most accessible introduction to design idea generation for ID and HCI. There are also numerous examples of prototyping strategies, ways in which to generate new ideas and demonstrations of participants actively involved in design idea generation. Some are presented later as case studies (Chapter 7) while many can be accessed or downloaded as videos (see the libraries of video material indicated in the Preface, under Sources of further information). Video material is especially useful in showing the reality of idea generation in action.

[25] *Buxton, B. (2007).* Sketching User Experiences *and Buxton, B. et al. (2010).* Sketching User Experiences: The Workbook.

Idea generation is an actively researched topic in the HCI, ID, UX and design communities, and articles, studies and investigations are constantly being presented at conferences, especially at the DIS (Designing Interactive Systems), NordiCHI (Nordic Conference on Human-Computer Interaction), UbiComp (Ubiquitous Computing), MobileHCI (Human-Computer Interaction with Mobile Devices and Services), PDC (Participatory Design Conference), Pervasive (Pervasive Computing), TEI (Tangible, Embedded, and Embodied Interaction) and UX (User Experience) conferences referred to in the Preface. Current work and work-in-progress is written up in magazines and journals. Some newer techniques may also be described on the various research community websites or on vendors' and computer consultancies' blogs and news feeds (see, again, Preface: Sources of further information). It is well worth keeping up-to-date with new developments by accessing the latest web-based information.

The process of idea generation is an active one, requiring practice. Some real implementation of storyboarding and early prototyping techniques may be carried out as coursework but students should all try the techniques for themselves, both as individuals and by working in groups. Some suggestions are made in the Exercises at the end of this chapter and there are many more in all of the recommended textbooks. The only way to gain experience and develop expertise is to try sketching and producing design ideas for yourself.

Summary

The goal of Interaction Design is to create products which are useable, useful and desirable. ID attempts to manage the complexity of interaction without losing the feel of a craft when creating 'great user experiences'. The design approach is one of focusing on users and their needs, and is derived both from design theory and from existing SE and HCI User-Centred Design models.

A reminder of your learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- demonstrate a realistic appreciation of ID processes and activities
- discuss the variety of approaches to ID and the range of design techniques and methodologies
- describe how and when such methods are used in a design activity
- describe what a metaphor is, and identify its importance
- demonstrate practice in drawing, sketching and designing paper prototypes.

Exercises

1. What is User-Centred Design (UCD)? What specific knowledge is required to engage in UCD?
2. Assume you are a member of a large software development team.
   i) How could you involve users at an early stage in systems development?
   ii) What problems and attitudes might you encounter?
   iii) How would you get users practically involved, and when?
3. Describe the design approach of Participative or Participatory Design.
4. What new kinds of software support are available for interface designers? How could you envisage yourself using them in the future?
5. Find out about 3D printers and investigate how they are being used in ID and UX Design.
6. How could you as a designer make use of users' prior experience and existing knowledge?
7. Describe the different ways in which designers use mock-ups, storyboards and paper prototypes.
8. Investigate video scenarios and video prototypes. How and when might they be used? What kinds of user might they be most suited to?
9. Sketch out three or four different design ideas for the interface to a new mobile phone application of your own choice. Make one touch-screen based and another keypad-based. How do they differ?
10. Construct a physical mock-up, with any readily accessible and available materials, of a personal diary/appointment calendar system for a laptop or notebook computer. Show it to someone else (preferably not on your course) to get their reaction and feedback.

Sample examination question

The University has a policy that all students must have new iPads for a specific course. From the perspective of an Interaction Designer, how would you apply a recognised design method to the process of designing a suitable application for students to use to access local university information and to participate in a wide range of class and campus activities? Provide mock-ups and paper-based prototypes or sketches of your design solution. Briefly critique your design.

Further reading

For background reading and inspiration for Interactive Design from a number of classic HCI perspectives

Blythe, M., et al. *Funology: from usability to enjoyment*. (2003)

Moggridge, B. *Designing interactions*. (2007).

Norman, D. *The design of everyday things*. (2002).

Norman, D. *The design of future things*. (2007).

Norman, D. *Living with complexity*. (2010).

Winograd, T. *Bringing design to software*. (1996).

New and newly-updated titles and synthesis lectures on design issues, UX and Interaction Design

Goodwin, K. *Designing for the digital age*. (2009).

Kuniavsky, M. *Smart things: ubiquitous computing user experience design*. (2010).

Hassenzahl, M. *Experience design: technology for all the right reasons*. (2010).

Saffer, D. *Designing for interaction*. (2009).

Sutcliffe, A. *Designing for user engagement: aesthetic and attractive user interfaces*. (2009).

Wilson, C. (ed.) *User experience re-mastered*. (2009).

Wright, P. and J. McCarthy *Experience-centred design: designers, users, and communities in dialogue*. (2009).

Appendix 3
Chapter summaries

Preface

About this course unit » Course aims, learning objectives and outcomes » The Study Guide » Recommended texts and supporting resources » Sources of further information » Acronyms

Learning outcomes

By the end of this half unit, and having completed the relevant readings and activities, you should be able to:

- gain a historical perspective of the field of HCI and its relationship with software engineering, psychology and ergonomics
- understand what Interaction Design is and be able to think critically about design
- appreciate HCI theory, practice and a set of design approaches
- understand the concept of usability and techniques of usability evaluation
- make use of usability concepts in appreciating, using and building interactive solutions for a range of applications
- criticise and improve poor interface designs, and so develop the skills to design and evaluate interfaces
- demonstrate awareness of new application areas and up-and-coming advanced technologies in order to better understand the potential of new technologies and techniques in the design of future interactive systems and applications.

Chapter 1: Introduction to HCI and Interaction Design

Introduction » Definitions » Why study user interaction » Early HCI » Why it changed » HCI in the 1970s and 1980s » The situation today » Summary

Summary

The scope of HCI has progressed from ergonomics/human factors to human cognition and psychology; from effects on workflow and communication to effects on culture and society. We now investigate the relationships between people that computers and computer networks enable, such as patterns of behaviour between people and within social groups, and study the digital artefacts which shape aspects of our everyday lives.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- understand the definitions of HCI and ID and associated terminology
- demonstrate knowledge of the historical development of the discipline
- define and describe common HCI terms
- describe how future developments in interaction, design and technology link to the history of the study of ID and HCI.

Chapter 2: Interaction and design approaches

Introduction » What is design? » Some principles of design » Problem space and design space » Design methodologies and approaches » The design activity » Idea generation » Summary

Summary

The goal of Interaction Design is to create products which are useable, useful and desirable. ID attempts to manage the complexity of interaction without losing the feel of a craft when creating ‘great user experiences’. The design approach is one of focusing on users and their needs, and is derived both from design theory and from existing SE and HCI User-Centred Design models.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- demonstrate a realistic appreciation of ID processes and activities
- discuss the variety of approaches to ID and the range of design techniques and methodologies
- describe how and when such methods are used in a design activity
- describe what a metaphor is, and identify its importance
- demonstrate practice in drawing, sketching and designing paper prototypes.

Chapter 3: Techniques for Interaction Design Requirements

Introduction » Task Analysis » Data collection » Qualitative data » User narratives and prototyping » Requirements gathering » Summary

Summary

Interaction Design is about understanding user needs, taking into account what people are good and bad at, and what can help them best in the way they currently do things. Determining user needs and translating them into system requirements is achieved primarily in user-centred methods by listening to what people want and getting them involved in the design activity. Different data gathering techniques are used to capture and discover user needs, but data interpretation and analysis requires expertise and often an appreciation of the embedded social aspects of work. Task Analysis identifies and describes tasks and user activities, and their inter-relationships; task scenarios, user narratives and prototypes can all be used both to generate designs and to see how users might interact with proposed design alternatives.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- describe a range of user-focused requirements elicitation techniques
- use a number of techniques for obtaining design input from an end-user population
- understand different types of data (qualitative and quantitative)
- describe which techniques to use in which situations
- demonstrate knowledge of the use of prototyping techniques and distinctions between those techniques.

Chapter 4: Usability

Introduction » Usability » International standards » The Usability Profession » Summary

Summary

Quantifiable elements of usability can be defined through the measurement of efficiency, effectiveness and satisfaction factors. Such usability metrics can be used for comparative testing of systems or interfaces through the definition of usability goals. Standardisation of terminology, approach and techniques have had a great impact within the field of usability testing, on usability design and engineering approaches and on the Usability Profession itself.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- define and describe usability and usability goals
- discuss how to assess usability and know what usability evaluation entails
- describe the business benefits of usability
- demonstrate a good appreciation of international standards concerned with usability
- describe what a Usability Practitioner does.

Chapter 5: Evaluation and usability assessment techniques

Introduction » Informal techniques » Heuristic Evaluation » Cognitive Walkthrough » Formal techniques » WoZ » Experimental evaluation » Choosing a technique » Professional input » Summary

Summary

Evaluation occurs at different stages of the design process and assesses different aspects of design. At the early design specification stage, it is **formative**, before implementation is finalised. At the later design implementation stage, after a runnable program has been produced, with a full prototype or the final system, it is **summative**. By its very nature, however, all evaluation is highly iterative. Evaluation paradigms for testing systems and interfaces can be informally classified:

- 'quick and dirty' – fast, informal data
- concerned with usability testing – carefully controlled
- field studies – situated in natural conditions
- predictive – actual users not required.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- describe specific usability evaluation techniques
- undertake simple assessments and interface design evaluations
- describe how to choose between evaluation methods and techniques.

Chapter 6: Designing for different users

Introduction » Culture and universal usability » Inclusive interaction » Accessibility » Summary

Summary

Designing interactions, interfaces and applications for new technologies is not just about designing for a set of given users or for generic populations. When designing for a broad range of abilities many factors must be taken into account to enable Universal Usability and appropriate cross-cultural designs to cater for cultural diversity. Inclusive Interaction requires a good understanding of the needs of all potential users – children, the elderly and those with special needs and impairments or disabilities. Enabling technology for accessibility leads to Assistive Technologies. Mobile and website design must increasingly conform to legal accessibility requirements and guidelines.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- understand and be able to describe the concepts of Universal Usability and Universal Design
- describe the needs and physical requirements of differing user populations
- discuss how changes in user populations can impact on Interaction Design.

Chapter 7: Design case studies

Introduction » hcibook » ID2 » HFRG » Equator » UPA » GATECH » Summary

Summary

A list of case studies has been provided and short descriptions, indicating the area of investigation and the design issues and techniques applied as solutions, have been given.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- demonstrate an understanding of what a case study in HCI and ID is, and what it entails
- demonstrate knowledge from reading about and studying the selected and recommended case studies
- obtain practice and experience in carrying out small prototyping, design and test exercises.

Chapter 8: Interaction Design and new technologies

Introduction » Tangible Interaction » Affective Interaction » Virtual Environments » Summary

Summary

Tangible Interaction concentrates on the haptic sense and tangibility – interaction is embedded in real spaces – whilst Whole-Body Interaction involves physical bodily interaction or movement related to physical objects. This type of interaction tries to embody some of the richness of interaction we have with physical devices into interaction with digital content. One way in which to achieve this is by tangible devices, or by embedding electronic sensors in everyday items such as clothing. Affective Interaction can improve both human-to-computer and human-to-human communications. Social robots or agents that can demonstrate affective reactions will be a feature of future applications, some in Immersive Virtual Environments. Designing VEs means understanding the concepts of immersion, engagement, context-awareness and presence to allow users to fully participate in a virtual world. Augmented interaction has myriad future uses which will give rise to new ways of interacting with computers and technology.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- describe some of the new and emerging technologies that will impact on Interaction Design
- discuss what some new user requirements might be
- decribe how to design for new technologies, needs and applications.

Chapter 9: Real world interactions

Introduction » Ubiquitous Computing » Seductive Interfaces and Flow » Ambient Intelligence » The Internet of Things » Impacts » Relevance to ID and HCI » Summary

Summary

Various trends have emerged which envisage people surrounded by computers embedded in everyday objects which would recognise and respond to individuals in a seamless, unobtrusive and invisible fashion. Ambient Intelligence emphasises human-centred computing in the convergence of innovations in three key technologies. Implementation is being carried out by commercial companies developing internet-enabled devices whilst research by international laboratories and longer-term legislative issues are being considered by governments. It is, however, a world of potential problems in identity, trust and social engineering.

Learning outcomes

By the end of this chapter, and having completed the relevant readings and activities, you should be able to:

- demonstrate an understanding of social aspects and societal impacts of new computer technologies
- understand and be able to define the concepts of Ubiquitous Computing, Ambient Intelligence and The Internet of Things
- discuss issues such as personal privacy and online trust in human-computer interactions
- describe the design challenges posed by new interaction paradigms.